Literature search report – run 20260828T100120

scitech-librarian 3.2

August 28, 2026

Generated by scitech-librarian 3.1 (report level: intermediate). Every number below is reproducible from the archived run directory `20260828T100120`.

Item	Value
Run started	2026-08-28 10:01:20
Duration	128 s
Query file	queries.example.json
Blocks	PSC, MLP, NOV, QC
Backends / sources	openalex, arxiv, inspire, semanticscholar, ads, scopus
Mode	full fetch, up to 300 records per block and backend
Non-curated venue filter	on
Open-access lookup	not run
Journal metrics	none (journals.py fetch)
Filters	none
Interrupted	no

Search strategy

Each block is one structural query – a conjunction of synonym groups, (a OR b) AND (c OR d) – rendered into every backend's native grammar. The strings below are exactly what was sent (PRISMA-S item 8).

Block PSC: Perovskite solar cell degradation under humidity

Purpose: grounding block – expect thousands of hits; use it to sanity-check that every backend is reachable and the counts are plausible

```
(perovskite solar cell OR halide perovskite photovoltaic) AND (degradation OR stability OR encapsulation)
 AND (humidity OR moisture OR water ingress)
```

Backend	Query string sent
openalex	("perovskite solar cell" OR "halide perovskite photovoltaic") AND (degradation OR stability OR encapsulation) AND (humidity OR moisture OR "water ingress")
arxiv	(all:"perovskite solar cell" OR all:"halide perovskite photovoltaic") AND (all:"degradation" OR all:"stability" OR all:"encapsulation")
inspire	("perovskite solar cell" or "halide perovskite photovoltaic") and ("degradation" or "stability" or "encapsulation") and ("humidity" or "moisture" or "water ingress")
semanticscholar	("perovskite solar cell" | "halide perovskite photovoltaic") + ("degradation" | "stability" | "encapsulation") + ("humidity" | "moisture" | "water ingress")
ads	(abs:"perovskite solar cell" OR abs:"halide perovskite photovoltaic") AND (abs:"degradation" OR abs:"stability" OR abs:"encapsulation") AND (abs:"humidity" OR abs:"moisture" OR abs:"water ingress")
scopus	TITLE-ABS-KEY("perovskite solar cell" OR "halide perovskite photovoltaic") AND TITLE-ABS-KEY(degradation OR stability OR encapsulation) AND TITLE-ABS-KEY(humidity OR moisture OR "water ingress")

Block MLP: Machine-learned interatomic potentials for amorphous materials

Purpose: a narrower intersection – expect hundreds; good for testing record fetch and RIS export

```
(machine learning potential OR machine-learned interatomic potential OR neural network potential) AND
(amorphous OR glass OR disordered solid) AND (molecular dynamics OR structure prediction)
```

arXiv receives groups [0, 1] only (nested-boolean limitation).

Backend	Query string sent
openalex	("machine learning potential" OR "machine-learned interatomic potential" OR "neural network potential") AND (amorphous OR glass OR "disordered solid") AND ("molecular dynamics" OR "structure prediction")
arxiv	(all:"machine learning potential" OR all:"machine-learned interatomic potential" OR all:"neural network potential") AND (all:"amorphous" OR all:"glass" OR all:"disordered solid")
inspire	("machine learning potential" or "machine-learned interatomic potential" or "neural network potential") and ("amorphous" or "glass" or "disordered solid") and ("molecular dynamics" or "structure prediction")
semanticscholar	("machine learning potential" | "machine-learned interatomic potential" | "neural network potential") + ("amorphous" | "glass" | "disordered solid") + ("molecular dynamics" | "structure prediction")
ads	(abs:"machine learning potential" OR abs:"machine-learned interatomic potential" OR abs:"neural network potential") AND (abs:"amorphous" OR abs:"glass" OR abs:"disordered solid") AND (abs:"molecular dynamics" OR abs:"structure prediction")
scopus	TITLE-ABS-KEY("machine learning potential" OR "machine-learned interatomic potential" OR "neural network potential") AND TITLE-ABS-KEY(amorphous OR glass OR "disordered solid") AND TITLE-ABS-KEY("molecular dynamics" OR "structure prediction")

Block NOV: EXAMPLE NOVELTY CHECK: origami metamaterials as topological acoustic pumps

Purpose: novelty-check pattern – a SMALL number is the good outcome; read every hit by hand before claiming a gap

```
(origami OR kirigami) AND (acoustic metamaterial OR phononic crystal) AND (topological pumping OR
Thouless pump OR edge state)
```

Backend	Query string sent
openalex	(origami OR kirigami) AND ("acoustic metamaterial" OR "phononic crystal") AND ("topological pumping" OR "Thouless pump" OR "edge state")
arxiv	(all:"origami" OR all:"kirigami") AND (all:"acoustic metamaterial" OR all:"phononic crystal")
inspire	("origami" or "kirigami") and ("acoustic metamaterial" or "phononic crystal") and ("topological pumping" or "Thouless pump" or "edge state")
semanticscholar	("origami" | "kirigami") + ("acoustic metamaterial" | "phononic crystal") + ("topological pumping" | "Thouless pump" | "edge state")
ads	(abs:"origami" OR abs:"kirigami") AND (abs:"acoustic metamaterial" OR abs:"phononic crystal") AND (abs:"topological pumping" OR abs:"Thouless pump" OR abs:"edge state")
scopus	TITLE-ABS-KEY(origami OR kirigami) AND TITLE-ABS-KEY("acoustic metamaterial" OR "phononic crystal") AND TITLE-ABS-KEY("topological pumping" OR "Thouless pump" OR "edge state")

Block QC: Quasicrystal photonics (arXiv-tuned example)

Purpose: demonstrates arxiv_groups: arXiv chokes on deeply nested booleans, so name the two groups it should get

```
(quasicrystal OR quasiperiodic lattice) AND (photonic OR waveguide OR optical lattice) AND (localization
OR critical states OR fractal spectrum)
```

arXiv receives groups [0, 2] only (nested-boolean limitation).

Backend	Query string sent
openalex	(quasicrystal OR "quasiperiodic lattice") AND (photonic OR waveguide OR "optical lattice") AND (localization OR "critical states" OR "fractal spectrum")
arxiv	(all:"quasicrystal" OR all:"quasiperiodic lattice") AND (all:"localization" OR all:"critical states" OR all:"fractal spectrum")
inspire	("quasicrystal" or "quasiperiodic lattice") and ("photonic" or "waveguide" or "optical lattice") and ("localization" or "critical states" or "fractal spectrum")
semanticscholar	("quasicrystal" | "quasiperiodic lattice") + ("photonic" | "waveguide" | "optical lattice") + ("localization" | "critical states" | "fractal spectrum")
ads	(abs:"quasicrystal" OR abs:"quasiperiodic lattice") AND (abs:"photonic" OR abs:"waveguide" OR abs:"optical lattice") AND (abs:"localization" OR abs:"critical states" OR abs:"fractal spectrum")
scopus	TITLE-ABS-KEY(quasicrystal OR "quasiperiodic lattice") AND TITLE-ABS-KEY(photonic OR waveguide OR "optical lattice") AND TITLE-ABS-KEY(localization OR "critical states" OR "fractal spectrum")

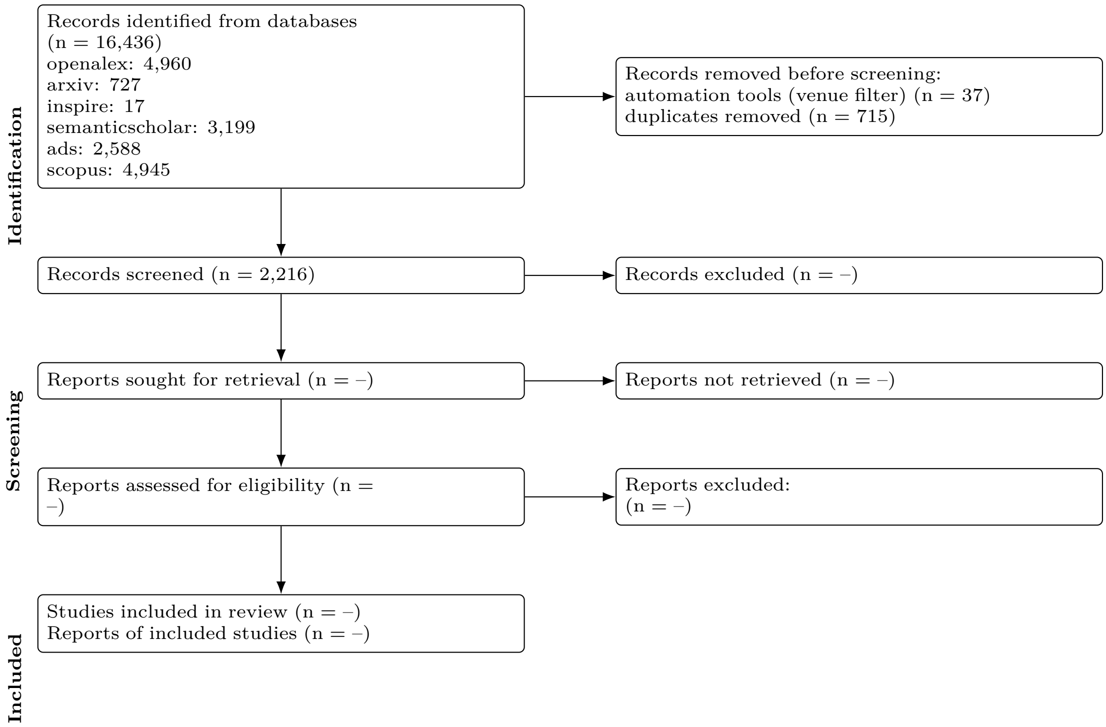
Results summary

Block	openalex	arxiv	inspire	semanticscholar	ads	scopus	Identified	Retrieved	Unique
PSC	4567	175	0	2968	2179	4731	14,620	1275	1163
MLP	239	138	0	109	203	149	838	802	444
NOV	0	0	0	0	0	0	0	0	0
QC	154	414	17	122	206	65	978	854	609

Block	openalex	arxiv	inspire	semanticscholar	ads	scopus	Identified	Retrieved	Unique
Total	4,960	727	17	3,199	2,588	4,945	16,436	2931	2216

Identified = database hit counts (not comparable across backends: proximity operators are dropped and stemming differs). Retrieved = records actually downloaded after the venue filter, capped by `--limit`. Unique = after DOI/title deduplication across all sources.

PRISMA 2020 flow



Stage	n
Records identified from databases	16,436
openalex	4,960
arxiv	727
inspire	17
semanticscholar	3,199
ads	2,588
scopus	4,945
Records retrieved (downloaded / ingested)	2,968
Removed before screening: automation (non-curated venues)	37
Removed before screening: duplicates	715
Records to screen (unique)	2,216
Records screened	2,216 (assumed = unique)
Records excluded at screening	–
Reports sought for retrieval	–
Reports not retrieved	–
Reports assessed for eligibility	–
Studies included	–
Reports of included studies	–

Automation stages are computed from the data; '–' marks manual stages not yet recorded in prisma.json. Note that 'identified' counts hits reported by each database while 'retrieved' is what was downloaded within `--limit`, so the two differ on large blocks.

PRISMA-S search-reporting checklist

Item	Requirement	This search
1	Database name	openalex, arxiv, inspire, semanticscholar, ads, scopus (documented public APIs)
2	Multi-database searching	6 databases, one structural query per block rendered into each native grammar; see Search strategy
3	Study registries	not applicable
4	Online resources and browsing	none recorded
5	Citation searching	not performed

Item	Requirement	This search
6	Contacts	none recorded
7	Other methods	none
8	Full search strategies	reported verbatim per backend under Search strategy; archived in queries.json
9	Limits and restrictions	record download capped at 300 per block and backend, most-cited first; no date, language or document-type limits applied
10	Search filters	venue filter on: records from non-curated repositories (Zenodo, Figshare, SSRN...) removed
11	Prior work	none
12	Updates	2 earlier run(s) archived; counts tracked in counts_history.csv
13	Dates of searches	searched on 2026-08-28 10:01:20
14	Peer review	none
15	Total records	16,436 identified from databases; 2,968 retrieved; 2,216 unique
16	Deduplication	exact DOI match, else first 90 characters of the lower-cased title; 715 duplicates removed

Records

Deduplicated across sources, sorted by cited.

Block PSC (1163 unique)

Title	Authors	Year	Venue	Cited	DOI
Incorporation of rubidium cations into perovskite solar cells improves photovoltaic performance	Michael Saliba; Taisuke Matsui; Konrad Domanski et al.	2016	Science	3691	10.1126/science.aah5557
High-efficiency two-dimensional Ruddlesden–Popper perovskite solar cells	Hsinhan Tsai; Wanyi Nie; Jean-Christophe Blancon et al.	2016	Nature	3360	10.1038/nature18306
Efficient, stable and scalable perovskite solar cells using poly(3-hexylthiophene)	Eui Hyuk Jung; Nam Joong Jeon; Eun Young Park et al.	2019	Nature	2318	10.1038/s41586-019-1036-3
A Layered Hybrid Perovskite Solar-Cell Absorber with Enhanced Moisture Stability	Ian C. P. Smith; Eric T. Hoke; Diego Solís-Ibarra et al.	2014	Angewandte Chemie International Edition	2006	10.1002/anie.201406466
Review of recent progress in chemical stability of perovskite solar cells	Guangda Niu; Xudong Guo; Liduo Wang	2014	Journal of Materials Chemistry A	1942	10.1039/c4ta04994b
Understanding Degradation Mechanisms and Improving Stability of Perovskite Photovoltaics	Caleb C. Boyd; Rongrong Cheacharoen; Tomas Leijtens et al.	2018	Chemical Reviews	1767	10.1021/acs.chemrev.8b00336
Formamidinium and Cesium Hybridization for Photo- and Moisture-Stable Perovskite Solar Cell	Jin-Wook Lee; Deok-Hwan Kim; Hui-Seon Kim et al.	2015	Advanced Energy Materials	1661	10.1002/aenm.201501310
23.6%-efficient monolithic perovskite/silicon tandem solar cells with improved stability	Kevin A. Bush; Axel F. Palmstrom; Zhengshan J. Yu et al.	2017	Nature Energy	1533	10.1038/nenergy.2017.9
Investigation of CH 3 NH 3 PbI 3 Degradation Rates and Mechanisms in Controlled Humidity Environments Using in Situ Techniques	Jinli Yang; Braden D. Siempelkamp; Dianyi Liu et al.	2015	ACS Nano	1342	10.1021/nn506864k
Carbon Nanotube/Polymer Composites as a Highly Stable Hole Collection Layer in Perovskite Solar Cells	Severin N. Habisreutinger; Tomas Leijtens; Giles E. Eperon et al.	2014	Nano Letters	1198	10.1021/nl501982b
Improved performance and stability of perovskite solar cells by crystal crosslinking with alkylphosphonic acid -ammonium chlorides	Xiong Li; M. Ibrahim Dar; Chenyi Yi et al.	2015	Nature Chemistry	1176	10.1038/nchem.2324
All-Inorganic Perovskite Solar Cells	Jia Liang; Caixing Wang; Yanrong Wang et al.	2016	Journal of the American Chemical Society	1091	10.1021/jacs.6b10227
Study on the stability of CH 3 NH 3 PbI 3 films and the effect of post-modification by aluminum oxide in all-solid-state hybrid solar cells	Guangda Niu; Wenzhe Li; Fanqi Meng et al.	2013	Journal of Materials Chemistry A	1067	10.1039/c3ta13606j
Organometallic-functionalized interfaces for highly efficient inverted perovskite solar cells	Zhen Li; Bo Li; Xin Wu et al.	2022	Science	1031	10.1126/science.abm8566
Degradation observations of encapsulated planar CH 3 NH 3 PbI 3 perovskite solar cells at high temperatures and humidity	Yu Han; S. Meyer; Yasmina Dkhissi et al.	2015	Journal of Materials Chemistry A	997	10.1039/c5ta00358j
Stability of perovskite solar cells	Dian Wang; Matthew Wright; Naveen Kumar Elumalai et al.	2016	Solar Energy Materials and Solar Cells	992	10.1016/j.solmat.2015.12.025

Title	Authors	Year	Venue	Cited	DOI
Strained hybrid perovskite thin films and their impact on the intrinsic stability of perovskite solar cells	Jingjing Zhao; Yehao Deng; Haotong Wei et al.	2017	Science Advances	965	10.1126/sciadv.aao5616
Scalable fabrication and coating methods for perovskite solar cells and solar modules	Nam-Gyu Park; Kai Zhu	2020	Nature Reviews Materials	902	10.1038/s41578-019-0176-2
Improving efficiency and stability of perovskite solar cells with photocurable fluoropolymers	Federico Bella; Gianmarco Griffini; Juan-Pablo Correa-Baena et al.	2016	Science	866	10.1126/science.aah4046
Thermochromic halide perovskite solar cells	Jia Lin; Minliang Lai; Letian Dou et al.	2018	Nature Materials	854	10.1038/s41563-017-0006-0
Damp heat–stable perovskite solar cells with tailored-dimensionality 2D/3D heterojunctions	Randi Azmi; Esma Ugur; Akmaral Seitkhan et al.	2022	Science	843	10.1126/science.abm5784
Tailored interfaces of unencapsulated perovskite solar cells for >1,000 hour operational stability	Jeffrey A. Christians; Philip Schulz; Jonathan S. Tinkham et al.	2018	Nature Energy	826	10.1038/s41560-017-0067-y
Stable High-Performance Perovskite Solar Cells via Grain Boundary Passivation	Tianqi Niu; Jing Lü; Rahim Munir et al.	2018	Advanced Materials	800	10.1002/adma.201706576
In Situ Growth of 2D Perovskite Capping Layer for Stable and Efficient Perovskite Solar Cells	Peng Chen; Yang Bai; Songcan Wang et al.	2018	Advanced Functional Materials	772	10.1002/adfm.201706923
2D perovskite stabilized phase-pure formamidinium perovskite solar cells	Jin-Wook Lee; Zhenghong Dai; Tae-Hee Han et al.	2018	Nature Communications	747	10.1038/s41467-018-05454-4
Suppression of atomic vacancies via incorporation of isovalent small ions to increase the stability of halide perovskite solar cells in ambient air	Makhsud I. Saidaminov; Junghwan Kim; Ankit Jain et al.	2018	Nature Energy	745	10.1038/s41560-018-0192-2
A Layered Hybrid Perovskite Solar-Cell Absorber with Enhanced Moisture Stability	Ian C. P. Smith; Eric T. Hoke; Diego Solís-Ibarra et al.	2014	Angewandte Chemie	692	10.1002/ange.201406466
Stable high efficiency two-dimensional perovskite solar cells via cesium doping	Xu Zhang; Xiaodong Ren; Bin Liu et al.	2017	Energy & Environmental Science	675	10.1039/c7ee01145h
A polymer scaffold for self-healing perovskite solar cells	Yicheng Zhao; Jing Wei; Heng Li et al.	2016	Nature Communications	666	10.1038/ncomms10228
Accelerated degradation of methylammonium lead iodide perovskites induced by exposure to iodine vapour	Shenghao Wang; Yan Jiang; Emilio J. Juárez-Pérez et al.	2016	Nature Energy	657	10.1038/nenergy.2016.195
All-Inorganic CsPbX 3 Perovskite Solar Cells: Progress and Prospects	Jingru Zhang; Gary Hodes; Zhiwen Jin et al.	2019	Angewandte Chemie International Edition	611	10.1002/anie.201901081
Intact 2D/3D halide junction perovskite solar cells via solid-phase in-plane growth	Yeoun-Woo Jang; Seungmin Lee; Kyung Mun Yeom et al.	2021	Nature Energy	606	10.1038/s41560-020-00749-7
Trapped charge-driven degradation of perovskite solar cells	Namyoung Ahn; Kwisung Kwak; Min Seok Jang et al.	2016	Nature Communications	601	10.1038/ncomms13422
Stability of Perovskite Solar Cells: A Prospective on the Substitution of the A Cation and X Anion	Ze Wang; Zejiao Shi; Taotao Li et al.	2016	Angewandte Chemie International Edition	595	10.1002/anie.201603694
Efficient and stable perovskite solar cells prepared in ambient air irrespective of the humidity	Qidong Tai; Peng You; Hongqian Sang et al.	2016	Nature Communications	582	10.1038/ncomms11105
Efficient and stable Ruddlesden–Popper perovskite solar cell with tailored interlayer molecular interaction	Hui Ren; Shidong Yu; Lingfeng Chao et al.	2020	Nature Photonics	581	10.1038/s41566-019-0572-6
Quantum-size-tuned heterostructures enable efficient and stable inverted perovskite solar cells	Hao Chen; Sam Teale; Bin Chen et al.	2022	Nature Photonics	581	10.1038/s41566-022-00985-1
Perovskite-polymer composite cross-linker approach for highly-stable and efficient perovskite solar cells	Tae-Hee Han; Jin-Wook Lee; Chungseok Choi et al.	2019	Nature Communications	581	10.1038/s41467-019-08455-z
Stabilizing the Efficiency Beyond 20% with a Mixed Cation Perovskite Solar Cell Fabricated in Ambient Air under Controlled Humidity	Trilok Singh; Tsutomu Miyasaka	2017	Advanced Energy Materials	573	10.1002/aenm.201700677
Two-dimensional Ruddlesden–Popper layered perovskite solar cells based on phase-pure thin films	Chao Liang; Hao Gu; Yingdong Xia et al.	2020	Nature Energy	554	10.1038/s41560-020-00721-5
Low-loss contacts on textured substrates for inverted perovskite solar cells	So Min Park; Mingyang Wei; Nikolaos Lempesis et al.	2023	Nature	548	10.1038/s41586-023-06745-7

Title	Authors	Year	Venue	Cited	DOI
Multifunctional Chemical Linker Imidazoleacetic Acid Hydrochloride for 21% Efficient and Stable Planar Perovskite Solar Cells	Jiangzhao Chen; Xing Zhao; Seul-Gi Kim et al.	2019	Advanced Materials	534	10.1002/adma.201902902
A review on perovskite solar cells: Evolution of architecture, fabrication techniques, commercialization issues and status	Priyanka Roy; Numeshwar Kumar Sinha; Sanjay Tiwari et al.	2020	Solar Energy	530	10.1016/j.solener.2020.01.080
CsPb0.9Sn0.1IBr2 Based All-Inorganic Perovskite Solar Cells with Exceptional Efficiency and Stability	Jia Liang; Peiyang Zhao; Caixing Wang et al.	2017	Journal of the American Chemical Society	528	10.1021/jacs.7b07949
Functionalization of perovskite thin films with moisture-tolerant molecules	Shuang Yang; Yun Wang; Porun Liu et al.	2016	Nature Energy	527	10.1038/nenergy.2015.16
Surface Passivation Using 2D Perovskites toward Efficient and Stable Perovskite Solar Cells	Guangbao Wu; Rui Liang; Mingzheng Ge et al.	2021	Advanced Materials	524	10.1002/adma.202105635
Engineering Stress in Perovskite Solar Cells to Improve Stability	Nicholas Rolston; Kevin A. Bush; Adam D. Printz et al.	2018	Advanced Energy Materials	510	10.1002/aenm.201802139
Enhanced Efficiency and Stability of Inverted Perovskite Solar Cells Using Highly Crystalline SnO 2 Nanocrystals as the Robust Electron-Transporting Layer	Zonglong Zhu; Yang Bai; Xiao Liu et al.	2016	Advanced Materials	496	10.1002/adma.201600619
Interfacial Residual Stress Relaxation in Perovskite Solar Cells with Improved Stability	Hao Wang; Cheng Zhu; Lang Liu et al.	2019	Advanced Materials	495	10.1002/adma.201904408
Degradation mechanism of hybrid tin-based perovskite solar cells and the critical role of tin (IV) iodide	Luis Lanzetta; Thomas Webb; Nourdine Zibouche et al.	2021	Nature Communications	493	10.1038/s41467-021-22864-z
Precise Control of Crystal Growth for Highly Efficient CsPbI2Br Perovskite Solar Cells	Weijie Chen; Haiyang Chen; Guiying Xu et al.	2018	Joule	491	10.1016/j.joule.2018.10.011
Gas chromatography–mass spectrometry analyses of encapsulated stable perovskite solar cells	Lei Shi; Martin P. Bucknall; Trevor L. Young et al.	2020	Science	489	10.1126/science.aba2412
Engineering ligand reactivity enables high-temperature operation of stable perovskite solar cells	So Min Park; Mingyang Wei; Jian Xu et al.	2023	Science	482	10.1126/science.adi4107
Advances in hole transport materials engineering for stable and efficient perovskite solar cells	Zinab H. Bakr; Qamar Wali; Azhar Fakharuddin et al.	2017	Nano Energy	448	10.1016/j.nanoen.2017.02.025
Flexible Perovskite Solar Cells	Hyun Suk Jung; Gill Sang Han; Nam-Gyu Park et al.	2019	Joule	446	10.1016/j.joule.2019.07.023
Perovskite Solar Cell Stability in Humid Air: Partially Reversible Phase Transitions in the PbI 2 -CH 3 NH 3 I-H 2 O System	Zhaoning Song; Antonio Abate; Suneth C. Watthage et al.	2016	Advanced Energy Materials	435	10.1002/aenm.201600846
Is Excess PbI2 Beneficial for Perovskite Solar Cell Performance?	Fangzhou Liu; Qi Dong; Man Kwong Wong et al.	2016	Advanced Energy Materials	422	10.1002/aenm.201502206
Device stability of perovskite solar cells – A review	Muhammad Imran Asghar; Jun Zhang; H. Wang et al.	2017	Renewable and Sustainable Energy Reviews	418	10.1016/j.rser.2017.04.003
Multifunctional Enhancement for Highly Stable and Efficient Perovskite Solar Cells	Yuan Cai; Jian Cui; Ming Chen et al.	2020	Advanced Functional Materials	418	10.1002/adfm.202005776
Enhancing stability and efficiency of perovskite solar cells with crosslinkable silane-functionalized and doped fullerene	Yang Bai; Qingfeng Dong; Yuchuan Shao et al.	2016	Nature Communications	416	10.1038/ncomms12806
Tailored Amphiphilic Molecular Mitigators for Stable Perovskite Solar Cells with 23.5% Efficiency	Hongwei Zhu; Yuhang Liu; Felix T. Eickemeyer et al.	2020	Advanced Materials	416	10.1002/adma.201907757
Unveiling facet-dependent degradation and facet engineering for stable perovskite solar cells	Chunqing Ma; Felix T. Eickemeyer; Sun-Ho Lee et al.	2023	Science	414	10.1126/science.adf3349
2D/3D heterojunction engineering at the buried interface towards high-performance inverted methylammonium-free perovskite solar cells	Jing Li; Cong Zhang; Cheng Gong et al.	2023	Nature Energy	395	10.1038/s41560-023-01295-8

Title	Authors	Year	Venue	Cited	DOI
Orientation Regulation of Phenylethylammonium Cation Based 2D Perovskite Solar Cell with Efficiency Higher Than 11%	Xinqian Zhang; Gang Wu; Weifei Fu et al.	2018	Advanced Energy Materials	389	10.1002/aenm.201702498
Isomer-Pure Bis-PCBM-Assisted Crystal Engineering of Perovskite Solar Cells Showing Excellent Efficiency and Stability	Fei Zhang; Wenda Shi; Jingshan Luo et al.	2017	Advanced Materials	388	10.1002/adma.201606806
UV Degradation and Recovery of Perovskite Solar Cells	Sang-Won Lee; Seongtak Kim; Soohyun Bae et al.	2016	Scientific Reports	383	10.1038/srep38150
A chemically inert bismuth interlayer enhances long-term stability of inverted perovskite solar cells	Shaohang Wu; Rui Chen; Shasha Zhang et al.	2019	Nature Communications	378	10.1038/s41467-019-09167-0
Polymer Doping for High-Efficiency Perovskite Solar Cells with Improved Moisture Stability	Jiexuan Jiang; Qian Wang; Zhiwen Jin et al.	2017	Advanced Energy Materials	377	10.1002/aenm.201701757
Humidity-Induced Degradation via Grain Boundaries of HC(NH2)2PbI3 Planar Perovskite Solar Cells	Jae Sung Yun; Jincheol Kim; Trevor L. Young et al.	2018	Advanced Functional Materials	373	10.1002/adfm.201705363
Optimal Interfacial Engineering with Different Length of Alkylammonium Halide for Efficient and Stable Perovskite Solar Cells	Hyeonwoo Kim; Seungun Lee; Do Yoon Lee et al.	2019	Advanced Energy Materials	353	10.1002/aenm.201902740
Enhancing Stability of Perovskite Solar Cells to Moisture by the Facile Hydrophobic Passivation	Insung Hwang; Inyoung Jeong; Jinwoo Lee et al.	2015	ACS Applied Materials & Interfaces	352	10.1021/acsami.5b04490
Passivation of Grain Boundaries by Phenethylammonium in Formamidinium-Methylammonium Lead Halide Perovskite Solar Cells	Da Seul Lee; Jae Sung Yun; Jincheol Kim et al.	2018	ACS Energy Letters	349	10.1021/acsenergylett.8b00121
Material and Device Stability in Perovskite Solar Cells	Hui-Seon Kim; Ja-Young Seo; Nam-Gyu Park	2016	ChemSusChem	347	10.1002/cssc.201600915
Mixed Dimensional 2D/3D Hybrid Perovskite Absorbers: The Future of Perovskite Solar Cells?	Anurag Krishna; Sébastien Gottis; Mohammad Khaja Nazeeruddin et al.	2018	Advanced Functional Materials	347	10.1002/adfm.201806482
Mixed Cation FAxPEA1–xPbI3 with Enhanced Phase and Ambient Stability toward High-Performance Perovskite Solar Cells	Nan Li; Zonglong Zhu; Chu-Chen Chueh et al.	2016	Advanced Energy Materials	345	10.1002/aenm.201601307
Improvement of the humidity stability of organic–inorganic perovskite solar cells using ultrathin Al 2 O 3 layers prepared by atomic layer deposition	Xu Dong; Xiang Fang; Minghang Lv et al.	2015	Journal of Materials Chemistry A	343	10.1039/c4ta06128d
Encapsulation for long-term stability enhancement of perovskite solar cells	Fabio Matteocci; Lucio Cinà; Enrico Lamanna et al.	2016	Nano Energy	337	10.1016/j.nanoen.2016.09.041
Phase Pure 2D Perovskite for High-Performance 2D–3D Heterostructured Perovskite Solar Cells	Pengwei Li; Yiqiang Zhang; Chao Liang et al.	2018	Advanced Materials	336	10.1002/adma.201805323
Boosting the efficiency of carbon-based planar CsPbBr3 perovskite solar cells by a modified multistep spin-coating technique and interface engineering	Xingyue Liu; Xianhua Tan; Zhiyong Liu et al.	2018	Nano Energy	336	10.1016/j.nanoen.2018.11.053
High-Efficiency Perovskite Solar Cells with Imidazolium-Based Ionic Liquid for Surface Passivation and Charge Transport	Xuejie Zhu; Minyong Du; Jiangshan Feng et al.	2020	Angewandte Chemie International Edition	335	10.1002/anie.202010987
Enhanced Environmental Stability of Planar Heterojunction Perovskite Solar Cells Based on Blade-Coating	Jong H. Kim; Spencer T. Williams; Namchul Cho et al.	2015		335	10.1002/aenm.201401229
Mixed 3D–2D Passivation Treatment for Mixed-Cation Lead Mixed-Halide Perovskite Solar Cells for Higher Efficiency and Better Stability	Yongyoon Cho; Arman Mahboubi Soufiani; Jae Sung Yun et al.	2018	Advanced Energy Materials	334	10.1002/aenm.201703392
Role of 4- tert -Butylpyridine as a Hole Transport Layer Morphological Controller in Perovskite Solar Cells	Shen Wang; Mahsa Sina; Pritesh Parikh et al.	2016	Nano Letters	333	10.1021/acs.nanolett.6b02158
Room-Temperature Molten Salt for Facile Fabrication of Efficient and Stable Perovskite Solar Cells in Ambient Air	Lingfeng Chao; Yingdong Xia; Bixin Li et al.	2019	Chem	324	10.1016/j.chempr.2019.02.025

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Fundamental Understanding of Photocurrent Hysteresis in Perovskite Solar Cells	Pengyun Liu; Wei Wang; Shaomin Liu et al.	2019	Advanced Energy Materials	321	10.1002/aenm.201803017
Red-Carbon-Quantum-Dot-Doped SnO2 Composite with Enhanced Electron Mobility for Efficient and Stable Perovskite Solar Cells	Wei Hui; Yingguo Yang; Quan Xu et al.	2019	Advanced Materials	313	10.1002/adma.201906374
Reduction of bulk and surface defects in inverted methylammonium- and bromide-free formamidinium perovskite solar cells	Rui Chen; Jianan Wang; Zonghao Liu et al.	2023	Nature Energy	299	10.1038/s41560-023-01288-7
High-Performance Perovskite Solar Cells with Enhanced Environmental Stability Based on a (p-FC6H4C2H4NH3)2[PbI4] Capping Layer	Qin Zhou; Lusheng Liang; Junjie Hu et al.	2019	Advanced Energy Materials	295	10.1002/aenm.201802595
Stable and Efficient Organo-Metal Halide Hybrid Perovskite Solar Cells via -Conjugated Lewis Base Polymer Induced Trap Passivation and Charge Extraction	Pingli Qin; Guang Yang; Zhiwei Ren et al.	2018	Advanced Materials	293	10.1002/adma.201706126
Dion-Jacobson Phase 2D Layered Perovskites for Solar Cells with Ultrahigh Stability	Ahmad, Sajjad; Fu, Ping; Yu, Shuwen et al.	2019		292	10.1016/j.joule.2018.11.026
Unveiling the Role of tBP–LiTFSI Complexes in Perovskite Solar Cells	Shen Wang; Zihan Huang; Xuefeng Wang et al.	2018	Journal of the American Chemical Society	290	10.1021/jacs.8b09809
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Encapsulation of Organic and Perovskite Solar Cells: A Review	Ashraf Uddin; Mushfika Baishakhi Upama; Haimang Yi et al.	2019	Coatings	285	10.3390/coatings9020065
Metal Oxide Charge Transport Layers for Efficient and Stable Perovskite Solar Cells	Seong Sik Shin; Seon Joo Lee; Sang Il Seok	2019	Advanced Functional Materials	285	10.1002/adfm.201900455
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Fluorinated Low-Dimensional Ruddlesden–Popper Perovskite Solar Cells with over 17% Power Conversion Efficiency and Improved Stability	Jishan Shi; Yerun Gao; Xiang Gao et al.	2019	Advanced Materials	283	10.1002/adma.201901673
Influence of Electrode Interfaces on the Stability of Perovskite Solar Cells: Reduced Degradation Using MoOx/Al for Hole Collection	Erin M. Sanehira; Bertrand J. Tremolet de Villers; Philip Schulz et al.	2016	ACS Energy Letters	282	10.1021/acsenergylett.6b00013
Towards linking lab and field lifetimes of perovskite solar cells	Qi Jiang; Robert Tirawat; Ross A. Kerner et al.	2023	Nature	281	10.1038/s41586-023-06610-7
Highly Air-Stable Carbon-Based -CsPbI3 Perovskite Solar Cells with a Broadened Optical Spectrum	Sisi Xiang; Zhongheng Fu; Weiping Li et al.	2018	ACS Energy Letters	279	10.1021/acsenergylett.8b00820
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Interfacial Defect Passivation and Stress Release via Multi-Active-Site Ligand Anchoring Enables Efficient and Stable Methylammonium-Free Perovskite Solar Cells	Baibai Liu; Huan Bì; Dongmei He et al.	2021	ACS Energy Letters	276	10.1021/acsenergylett.1c00794
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Suppressed decomposition of organometal halide perovskites by impermeable electron-extraction layers in inverted solar cells	K. Brinkmann; Jie Zhao; Jie Zhao et al.	2017	Nature Communications	269	10.1038/ncomms13938
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Interface Modification by Ionic Liquid: A Promising Candidate for Indoor Light Harvesting and Stability Improvement of Planar Perovskite Solar Cells	Meng Li; Chao Zhao; Zhao-Kui Wang et al.	2018	Advanced Energy Materials	257	10.1002/aenm.201801509
Vertically Oriented 2D Layered Perovskite Solar Cells with Enhanced Efficiency and Good Stability	Xinqian Zhang; Gang Wu; Shida Yang et al.	2017	Small	255	10.1002/smll.201700611
A Review: Thermal Stability of Methylammonium Lead Halide Based Perovskite Solar Cells	Tanzila Tasnim Ava; Abdullah Al Mamun; Sylvain Marsillac et al.	2019	Applied Sciences	254	10.3390/app9010188
Multifunctional Phosphorus-Containing Lewis Acid and Base Passivation Enabling Efficient and Moisture-Stable Perovskite Solar Cells	Zhi Yang; Jinjuan Dou; Song Kou et al.	2020	Advanced Functional Materials	254	10.1002/adfm.201910710
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Intermolecular Exchange Boosts Efficiency of Air-Stable, Carbon-Based All-Inorganic Planar CsPbIBr2 Perovskite Solar Cells to Over 9%	Weidong Zhu; Qianni Zhang; Dazheng Chen et al.	2018	Advanced Energy Materials	251	10.1002/aenm.201802080
Interfacial defect passivation and stress release by multifunctional KPF6 modification for planar perovskite solar cells with enhanced efficiency and stability	Huan Bì; Baibai Liu; Dongmei He et al.	2021	Chemical Engineering Journal	250	10.1016/j.cej.2021.129375
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Inorganic Rubidium Cation as an Enhancer for Photovoltaic Performance and Moisture Stability of HC(NH2)2PbI3 Perovskite Solar Cells	Yun Hee Park; Inyoung Jeong; Seunghwan Bae et al.	2017	Advanced Functional Materials	248	10.1002/adfm.201605988
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Zwitterionic-Surfactant-Assisted Room-Temperature Coating of Efficient Perovskite Solar Cells	Kuan Liu; Qiong Liang; Minchao Qin et al.	2020	Joule	226	10.1016/j.joule.2020.09.011
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Compositional Engineering for Thermally Stable, Highly Efficient Perovskite Solar Cells Exceeding 20% Power Conversion Efficiency with 85 °C/85% 1000 h Stability	Taisuke Matsui; Teruaki Yamamoto; Takashi Nishihara et al.	2019	Advanced Materials	222	10.1002/adma.201806823
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Encapsulation of Perovskite Solar Cells for High Humidity Conditions	Qi Dong; Fangzhou Liu; Man Kwong Wong et al.	2016	ChemSusChem	219	10.1002/cssc.201600868
A Low-Temperature Thin-Film Encapsulation for Enhanced Stability of a Highly Efficient Perovskite Solar Cell	Young Il Lee; Nam Joong Jeon; Bong Jun Kim et al.	2017	Advanced Energy Materials	217	10.1002/aenm.201701928
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Enhancing Efficiency and Stability of Photovoltaic Cells by Using Perovskite/Zr-MOF Heterojunction Including Bilayer and Hybrid Structures	Chia-Chen Lee; Chih-I Chen; Yu-Te Liao et al.	2019	Advancement of science	212	10.1002/advs.201801715
Tailoring the Functionality of Organic Spacer Cations for Efficient and Stable Quasi-2D Perovskite Solar Cells	Weifei Fu; Hongbin Liu; Xueliang Shi et al.	2019	Advanced Functional Materials	211	10.1002/adfm.201900221

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Thermodynamically Self-Healing 1D–3D Hybrid Perovskite Solar Cells	Jiandong Fan; Yunping Ma; Cuiling Zhang et al.	2018	Advanced Energy Materials	207	10.1002/aenm.201703421
Synergistic Effects of Eu-MOF on Perovskite Solar Cells with Improved Stability	Jie Dou; Cheng Zhu; Hao Wang et al.	2021	Advanced Materials	206	10.1002/adma.202102947
Wafer-scale monolayer MoS 2 film integration for stable, efficient perovskite solar cells	Huachao Zai; Pengfei Yang; Jie Su et al.	2025	Science	206	10.1126/science.ado2351
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Continuous Grain-Boundary Functionalization for High-Efficiency Perovskite Solar Cells with Exceptional Stability	Yingxia Zong; Yuanyuan Zhou; Yi Zhang et al.	2018	Chem	204	10.1016/j.chempr.2018.03.005
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Review of Novel Passivation Techniques for Efficient and Stable Perovskite Solar Cells	Jincheol Kim; Anita Ho-Baillie; Shujuan Huang	2019	Solar RRL	185	10.1002/solr.201800302
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Enhanced Performance and Stability of 3D/2D Tin Perovskite Solar Cells Fabricated with a Sequential Solution Deposition	Efat Jokar; Po-Yuan Cheng; Chia-Yi Lin et al.	2021	ACS Energy Letters	161	10.1021/acsenergylett.0c02305
The Role of Charge Selective Contacts in Perovskite Solar Cell Stability	Bart Roose; Qiong Wang; Antonio Abate	2018	Advanced Energy Materials	155	10.1002/aenm.201803140
Over 24% efficient MA-free CsxFA1−xPbX3 perovskite solar cells	Siyang Wang; Liguo Tan; Junjie Zhou et al.	2022	Joule	155	10.1016/j.joule.2022.05.002
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A review of stability and progress in tin halide perovskite solar cell	Asim Aftab; Md. Imteyaz Ahmad	2021	Solar Energy	137	10.1016/j.solener.2020.12.065
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Cesiumlead based inorganic perovskite quantum-dots as interfacial layer for highly stable perovskite solar cells with exceeding 21% efficiency	Akin, Seckin; Altintas, Yemliha; Mutlugun, Evren et al.	2019		65	10.1016/j.nanoen.2019.03.091

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Chlorine-terminated MXene quantum dots for improving crystallinity and moisture stability in high-performance perovskite solar cells	Xiao Liu; Zhiang Zhang; Jinkun Jiang et al.	2021	Chemical Engineering Journal	64	10.1016/j.cej.2021.134382
Multifunctional Aminoglycoside Antibiotics Modified SnO2 Enabling High Efficiency and Mechanical Stability Perovskite Solar Cells	Tong Yan; Chenxi Zhang; Shiqi Li et al.	2023	Advanced Functional Materials	63	10.1002/adfm.202302336
A Review on Emerging Barrier Materials and Encapsulation Strategies for Flexible Perovskite and Organic Photovoltaics	Sutherland, Luke J.; Weerasinghe, Hasitha C.; Simon, George P.	2021		63	10.1002/aenm.202101383
Stability and Performance Enhancement of Perovskite Solar Cells: A Review	Maria Khalid; Tapas K. Mallick	2023	Energies	62	10.3390/en16104031
An integrated organic–inorganic hole transport layer for efficient and stable perovskite solar cells	Yaxiong Guo; Hongwei Lei; Liangbin Xiong et al.	2018		62	10.1039/C7TA09946K
Progress on the stability and encapsulation techniques of perovskite solar cells	Ling Xiang; Fangliang Gao; Yunxuan Cao et al.	2022	Organic Electronics	61	10.1016/j.orgel.2022.106515
Encapsulating perovskite solar cells for long-term stability and prevention of lead toxicity	Shahriyar Safat Dipta; Md. Arifur Rahim; Ashraf Uddin	2024	Applied Physics Reviews	61	10.1063/5.0197154
Air-processed mixed-cation Cs0.15FA0.85PbI3 planar perovskite solar cells derived from a PbI2–CsI–FAI intermediate complex	Xiuwen Xu; Chunqing Ma; Yue-Min Xie et al.	2018		61	10.1039/C8TA01049H
Bromine Doping as an Efficient Strategy to Reduce the Interfacial Defects in Hybrid Two-Dimensional/Three-Dimensional Stacking Perovskite Solar Cells.	Yanping Lv; Yantao Shi; Xuedan Song et al.	2018	ACS Applied Materials and Interfaces	61	10.1021/acsami.8b09461
Recent Progress on the Stability of Perovskite Solar Cells in a Humid Environment	Mengying Li; Haibo Li; Jing Fu et al.	2020	The Journal of Physical Chemistry C	60	10.1021/acs.jpcc.0c08019
Light enhanced moisture degradation of perovskite solar cell material CH 3 NH 3 PbI 3	Ying-Bo Lu; Wei-Yan Cong; ChengBo Guan et al.	2019	Journal of Materials Chemistry A	60	10.1039/c9ta10443g
Potassium iodide reduces the stability of triple-cation perovskite solar cells	Tarek I. Alanazi; Onkar S. Game; Joel A. Smith et al.	2020	RSC Advances	59	10.1039/d0ra07107b
Toward fast charge extraction in all-inorganic CsPbBr3 perovskite solar cells by setting intermediate energy levels	Guoqing Liao; Jialong Duan; Yuanyuan Zhao et al.	2018	Solar Energy	59	10.1016/J.SOLENER.2018.06.041
F-doping-Enhanced Carrier Transport in the SnO2/Perovskite Interface for High-Performance Perovskite Solar Cells.	Tianyuan Luo; Gang Ye; Xiayan Chen et al.	2022	ACS Applied Materials and Interfaces	59	10.1021/acsami.2c11390
Stable and scalable 3D-2D planar heterojunction perovskite solar cells via vapor deposition	Lin, Dongxu; Zhang, Tiankai; Wang, Jiming et al.	2019		58	10.1016/j.nanoen.2019.03.014
Precursor Engineering for Ambient-Compatible Antisolvent-Free Fabrication of High-Efficiency CsPbI₂Br Perovskite Solar Cells	Duan, Chenyang; Cui, Jian; Zhang, Miaomiao et al.	2020		58	10.1002/aenm.202000691
Enhanced moisture stability of metal halide perovskite solar cells based on sulfur–oleylamine surface modification	Yu Hou; Zi Ren Zhou; Tian Wen et al.	2018	Nanoscale Horizons	57	10.1039/c8nh00163d
Homeopathic Perovskite Solar Cells: Effect of Humidity during Fabrication on the Performance and Stability of the Device	Lidia Contreras-Bernal; Clara Aranda; Marta Vallés-Pelarda et al.	2018	The Journal of Physical Chemistry C	56	10.1021/acs.jpcc.8b01558
Nonionic Sc3N@C80 Dopant for Efficient and Stable Halide Perovskite Photovoltaics	Kai Wang; Xiaoyang Liu; Rong Huang et al.	2019	ACS Energy Letters	56	10.1021/acsenergylett.9b01042
Improving the moisture stability of perovskite solar cells by using PMMA/P3HT based hole-transport layers	Soumya Kundu; Timothy L. Kelly	2017	Materials Chemistry Frontiers	56	10.1039/c7qm00396j
Triamine-Based Aromatic Cation as a Novel Stabilizer for Efficient Perovskite Solar Cells	Jinhyun Kim; Alan Jiwan Yun; Bumjin Gil et al.	2019	Advanced Functional Materials	56	10.1002/adfm.201905190
Defect passivation by nontoxic biomaterial yields 21% efficiency perovskite solar cells	Shaobing Xiong; Tianyu Hao; Yuyun Sun et al.	2021		56	10.1016/j.jechem.2020.06.061

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C60 additive-assisted crystallization in CH3NH3Pb0.75Sn0.25I3 perovskite solar cells with high stability and efficiency.	Chong Liu; Wenzhe Li; Hongliang Li et al.	2017	Nanoscale	56	10.1039/c7nr03507a
Highly Efficient and Stable Perovskite Solar Cells Enabled by All-Crosslinked Charge-Transporting Layers	Zhu, Zonglong; Zhao, Dongbing; Chueh, Chu-Chen et al.	2018		56	10.1016/j.joule.2017.11.006
Moisture-tolerant supermolecule for the stability enhancement of organic–inorganic perovskite solar cells in ambient air	Dong Wei; Hao Huang; Peng Cui et al.	2018	Nanoscale	53	10.1039/c8nr07638c
Highest-Efficiency Flexible Perovskite Solar Module by Interface Engineering for Efficient Charge-Transfer	Dong Yang; Ruixia Yang; C. Zhang et al.	2023	Advances in Materials	53	10.1002/adma.202302484
Ge Incorporation to Stabilize Efficient Inorganic CsPbI₃ Perovskite Solar Cells	Meng, Fanqi; Yu, Bingcheng; Zhang, Qinghua et al.	2022		53	10.1002/aenm.202103690
Organic N-Type Molecule: Managing the Electronic States of Bulk Perovskite for High-Performance Photovoltaics	Haiyang Chen; Yu Zhan; Guiying Xu et al.	2020	Advanced Functional Materials	52	10.1002/adfm.202001788
Heterojunction Incorporating Perovskite and Microporous Metal–Organic Framework Nanocrystals for Efficient and Stable Solar Cells	Xuesong Zhou; Lele Qiu; Ruiqing Fan et al.	2020	Nano-Micro Letters	52	10.1007/s40820-020-00417-1
Multidentate Coordination Induced Crystal Growth Regulation and Trap Passivation Enables over 24% Efficiency in Perovskite Solar Cells	Runnan Yu; Guangzheng Wu; Rui Shi et al.	2022	Advanced Energy Materials	52	10.1002/aenm.202203127
The poly(styrene-co-acrylonitrile) polymer assisted preparation of high-performance inverted perovskite solar cells with efficiency exceeding 22%	Yang, Jiabao; Cao, Qi; He, Ziwei et al.	2021		52	10.1016/j.nanoen.2020.105731
Enhanced moisture stability of MAPbI3 perovskite solar cells through Barium doping	Jitendra Bahadur; Amir H. Ghahremani; Sunil Gupta et al.	2019	Solar Energy	51	10.1016/j.solener.2019.08.033
Light-Driven Dynamic Defect-Passivation for Efficient Inorganic Perovskite Solar Cells	Zhiteng Wang; Qiyong Chen; Huidong Xie et al.	2024	Advanced Functional Materials	51	10.1002/adfm.202416118
Anti-Solvent Assisted Crystallization Processed Methylammonium Bismuth Iodide Cuboids towards Highly Stable Lead-Free Perovskite Solar Cells	S. Mali; H. J. Kim; Do-Heyoung Kim et al.	2017		51	10.1002/SLCT.201700025
Thiocyanate Containing Two-Dimensional Cesium Lead Iodide Perovskite, Cs2PbI2(SCN)2: Characterization, Photovoltaic Application, and Degradation Mechanism.	Y. Numata; Y. Sanehira; R. Ishikawa et al.	2018	ACS Applied Materials and Interfaces	51	10.1021/acsami.8b15578
Interface Play between Perovskite and Hole Selective Layer on the Performance and Stability of Perovskite Solar Cells.	M. Salado; Jesús Idígoras; L. Calió et al.	2016	ACS Applied Materials and Interfaces	51	10.1021/acsami.6b12236
Sandwiched electrode buffer for efficient and stable perovskite solar cells with dual back surface fields	Zai, Huachao; Su, Jie; Zhu, Cheng et al.	2021		51	10.1016/j.joule.2021.06.001
Porous Lead Iodide Layer Promotes Organic Amine Salt Diffusion to Achieve High Performance p-i-n Flexible Perovskite Solar Cells	Sun, Qing; Duan, Shaocong; Liu, Gang et al.	2023		51	10.1002/aenm.202301259
Reduced Graphene Oxide Improves Moisture and Thermal Stability of Perovskite Solar Cells	Hui-Seon Kim; Bowen Yang; Minas M. Stylianakis et al.	2020	Cell Reports Physical Science	50	10.1016/j.xcrp.2020.100053
Large-area perovskite solar cells with CsxFA1−xPbI3−yBry thin films deposited by a vapor–solid reaction method	Long Luo; Yulong Zhang; Nianyao Chai et al.	2018		50	10.1039/C8TA06557H
Cerium-Oxide-Modified Anodes for Efficient and UV-Stable ZnO-Based Perovskite Solar Cells.	Ruiqian Meng; Xiaoxia Feng; Yiwei Yang et al.	2019	ACS Applied Materials and Interfaces	50	10.1021/acsami.9b01587

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Optimization of the Ag/PCBM interface by a rhodamine interlayer to enhance the efficiency and stability of perovskite solar cells.	John Ciro; Santiago Mesa; J. Uribe et al.	2017	Nanoscale	49	10.1039/c7nr01678f
Modifying SnO2 with Polyacrylamide to Enhance the Performance of Perovskite Solar Cells.	Hai Dong; Jilin Wang; Xingyu Li et al.	2022	ACS Applied Materials and Interfaces	49	10.1021/acsami.2c08662
High-Performance Multilayer Encapsulation for Perovskite Photovoltaics	Wong-Stringer, Michael; Game, Onkar S.; Smith, Joel A. et al.	2018		49	10.1002/aenm.201801234
A simple method to evaluate the effectiveness of encapsulation materials for perovskite solar cells	Timothy J. Wilderspin; Francesca De Rossi; Trystan Watson	2016	Solar Energy	48	10.1016/j.solener.2016.09.038
Pre-Embedded Multisite Chiral Molecules Realize Bottom-Up Multilayer Manipulation toward Stable and Efficient Perovskite Solar Cells	Qian Zhou; Baibai Liu; Yu Chen et al.	2024	Advanced Functional Materials	48	10.1002/adfm.202315064
Air-Induced High-Quality CH3NH3PbI3 Thin Film for Efficient Planar Heterojunction Perovskite Solar Cells	Chunhua Wang; Chujun Zhang; Sichao Tong et al.	2017		48	10.1021/ACS.JPCC.7B00981
Accelerated electron extraction and improved UV stability of TiO2 based perovskite solar cells by SnO2 based surface passivation	Fang Wan; Xincan Qiu; Hui Chen et al.	2018	Organic electronics	48	10.1016/J.ORGEL.2018.05.008
Mixed cations and mixed halide perovskite solar cell with lead thiocyanate additive for high efficiency and long-term moisture stability	Cai Ying; Shirong Wang; Mengna Sun et al.	2018		47	10.1016/J.ORGEL.2017.11.045
Extended Absorption Window and Improved Stability of Cesium-Based Triple-Cation Perovskite Solar Cells Passivated with Perfluorinated Organics	K. Salim; T. Koh; D. Bahulayan et al.	2018		47	10.1021/ACSENERGYLETT.8B00328
Simultaneously enhanced moisture tolerance and defect passivation of perovskite solar cells with cross-linked grain encapsulation	Ke Xiao; Qiaolei Han; Yuan Gao et al.	2020	Journal of Energy Chemistry	46	10.1016/j.jechem.2020.08.020
Protecting Perovskite Solar Cells against Moisture-Induced Degradation with Sputtered Inorganic Barrier Layers	Ramez Hosseinian Ahangharnejhad; Zhaoning Song; Tamanna Mariam et al.	2021	ACS Applied Energy Materials	46	10.1021/acsaem.1c00816
Synergistic Full-Scale Defect Passivation Enables High-Efficiency and Stable Perovskite Solar Cells	Wen, Haoxin; Zhang, Zhen; Guo, Yixuan et al.	2023		46	10.1002/aenm.202301813
Role of interface in stability of perovskite solar cells	Chris Manspeaker; Swaminathan Venkatesan; Alex Zakhidov et al.	2016	Current Opinion in Chemical Engineering	45	10.1016/j.coche.2016.08.013
Buried Interface Strategies with Covalent Organic Frameworks for High-Performance Inverted Perovskite Solar Cells	Shuai Yang; Jiandong He; Zhihui Chen et al.	2024	Advances in Materials	45	10.1002/adma.202408686
Highly flexible, robust, stable and high efficiency perovskite solar cells enabled by van der Waals epitaxy on mica substrate	Jia, Chunmei; Zhao, Xingyu; Lai, Yu-Hong et al.	2019		44	10.1016/j.nanoen.2019.03.053
Comprehensive passivation strategy for achieving inverted perovskite solar cells with efficiency exceeding 23% by trap passivation and ion constraint	Zhang, Fan; Ye, Shuai; Zhang, Hanhong et al.	2021		44	10.1016/j.nanoen.2021.106370
Efficient and stable planar all-inorganic perovskite solar cells based on high-quality CsPbBr₃ films with controllable morphology	Wan, Xiaojing; Yu, Ze; Tian, Wenming et al.	2020		43	10.1016/j.jechem.2019.10.017
Effective Surface Treatment for High-Performance Inverted CsPbI₂Br Perovskite Solar Cells with Efficiency of 15.92%	Fu, Sheng; Li, Xiaodong; Wan, Li et al.	2020		43	10.1007/s40820-020-00509-y
Defect passivation and electrical conductivity enhancement in perovskite solar cells using functionalized graphene quantum dots	Y. Rui; Zuoming Jin; Xinyi Fan et al.	2022	Materials Futures	41	10.1088/2752-5724/ac9707

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High-Mobility Hydrophobic Conjugated Polymer as Effective Interlayer for Air-Stable Efficient Perovskite Solar Cells	Xiao-Xin Gao; D. Xue; D. Gao et al.	2018	Solar RRL	41	10.1002/SOLR.201800232
Large-area, green solvent spray deposited nickel oxide films for scalable fabrication of triple-cation perovskite solar cells	Neetesh Kumar; H. Lee; Sun-Na Hwang et al.	2020		41	10.1039/c9ta13528f
Toward stable lead halide perovskite solar cells: A knob on the A/X sites components	Shurong Wang; Aili Wang; Feng Hao	2021	iScience	40	10.1016/j.isci.2021.103599
A Key 2D Intermediate Phase for Stable High-Efficiency CsPbI₂Br Perovskite Solar Cells	Yang, Shaomin; Wen, Jialun; Liu, Zhike et al.	2022		39	10.1002/aenm.202103019
Hysteresis and Instability Predicted in Moisture Degradation of Perovskite Solar Cells	Kelvin J. Xu; Ryan Taoran Wang; Fan Xu et al.	2020	ACS Applied Materials & Interfaces	38	10.1021/acsami.0c17323
Synergetic Excess PbI2 and Reduced Pb Leakage Management Strategy for 24.28% Efficient, Stable and Eco-Friendly Perovskite Solar Cells	Yuhong Zhang; Lin Xu; Yanjie Wu et al.	2023	Advanced Functional Materials	38	10.1002/adfm.202214102
Prolonged Lifetime of Perovskite Solar Cells Using a Moisture-Blocked and Temperature-Controlled Encapsulation System Comprising a Phase Change Material as a Cooling Agent	Nasibeh Mansour Rezaei Fumani; Farzaneh Arabpour Roghabadi; M. Alidaei et al.	2020	ACS Omega	37	10.1021/acsomega.9b03407
Dopant-Free Pyrrolopyrrole-Based (PPr) Polymeric Hole-Transporting Materials for Efficient Tin-Based Perovskite Solar Cells with Stability Over 6000 h	Chun-Hsiao Kuan; R. Balasaravanan; Shih-Min Hsu et al.	2023	Advances in Materials	37	10.1002/adma.202300681
Improved performance and air stability of planar perovskite solar cells via interfacial engineering using a fullerene amine interlayer	Xie, Jiangsheng; Yu, Xuegong; Sun, Xuan et al.	2016		37	10.1016/j.nanoen.2016.08.048
Morphological Insights into the Degradation of Perovskite Solar Cells under Light and Humidity	Kun Sun; Renjun Guo; Yuxin Liang et al.	2023	ACS Applied Materials & Interfaces	36	10.1021/acsami.3c05671
An efficient and hydrophobic molecular doping in perovskite solar cells	Junsheng Luo; Fang-Ru Lin; Jianxing Xia et al.	2021		36	10.1016/J.NANOEN.2021.105751
Recent progress on low dimensional perovskite solar cells	Lingfeng Chao; Ze Wang; Yingdong Xia et al.	2017	Journal of Energy Chemistry	36	10.1016/J.JECHEM.2017.10.013
Dopant-free hole transporting materials with supramolecular interactions and reverse diffusion for efficient and modular p-i-n perovskite solar cells	Rongming Xue; Moyao Zhang; Deying Luo et al.	2020	Science China Chemistry	36	10.1007/s11426-020-9741-1
Efficient, stable and flexible perovskite solar cells using two-step solution-processed SnO2 layers as electron-transport-material	Xincan Qiu; Bingchu Yang; Hui Chen et al.	2018	Organic electronics	36	10.1016/J.ORGEL.2018.04.010
Halide perovskite based on hydrophobic ionic liquid for stability improving and its application in high-efficient photovoltaic cell	Jue Wang; Xiaoxue Ye; Yanying Wang et al.	2019	Electrochimica Acta	36	10.1016/J.ELECTACTA.2019.02.071
Achieving Efficient and Stable Perovskite Solar Cells in Ambient Air Through Non-Halide Engineering	Wang, Zhen; Jin, Junjun; Zheng, Yapeng et al.	2021		36	10.1002/aenm.202102169
Perovskite Solar Cells in the Shadow: Understanding the Mechanism of Reverse-Bias Behavior toward Suppressed Reverse-Bias Breakdown and Reverse-Bias Induced Degradation	Wang, Chaofeng; Huang, Like; Zhou, Yike et al.	2023		36	10.1002/aenm.202203596
Efficient and stable mesoscopic perovskite solar cell in high humidity by localized Dion-Jacobson 2D-3D heterostructures	Li, Wenhui; Gu, Xiaoyu; Shan, Chengwei et al.	2022		35	10.1016/j.nanoen.2021.106666
Improvement of efficiency and stability of CuSCN-based inverted perovskite solar cells by post-treatment with potassium thiocyanate	Mei Lyu; Jiangzhao Chen; N. Park	2019	Journal of Solid State Chemistry	34	10.1016/J.JSSC.2018.10.014

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A crosslinked fullerene matrix doped with an ionic fullerene as a cathodic buffer layer toward high-performance and thermally stable polymer and organic metallohalide perovskite solar cells	Yi-Hsiang Chao; Y. Huang; Jen-Yun Chang et al.	2015		34	10.1039/C5TA05057J
p-Phenylenediaminium iodide capping agent enabled self-healing perovskite solar cell	P. Zardari; A. Rostami; H. Shekaari	2020	Scientific Reports	33	10.1038/s41598-020-76365-y
Locking Surface Dimensionality for Endurable Interface in Perovskite Photovoltaics	Xu Zhang; Yixin Luo; Xiaonan Wang et al.	2025	Carbon Energy	33	10.1002/cey2.718
Efficient and Moisture-Stable Inverted Perovskite Solar Cells via n-Type Small-Molecule-Assisted Surface Treatment	Ji A. Hong; Mingyu Jeong; Sujung Park et al.	2022	Advancement of science	32	10.1002/advs.202205127
Universal Bottom Contact Modification with Diverse 2D Spacers for High-Performance Inverted Perovskite Solar Cells	Jun Li; Lijian Zuo; Haotian Wu et al.	2021	Advanced Functional Materials	32	10.1002/adfm.202104036
Improved Moisture Stability of Perovskite Solar Cells Using N719 Dye Molecules	Minghua Zhang; Meiqian Tai; Xin Li et al.	2019	Solar RRL	32	10.1002/solr.201900345
Supramolecular Aza Crown Ether Modulator for Efficient and Stable Perovskite Solar Cells	Xia Chen; Chunyan Deng; Jihuai Wu et al.	2024	Advanced Functional Materials	32	10.1002/adfm.202311527
Alkaline-earth bis(trifluoromethanesulfonimide) additives for efficient and stable perovskite solar cells	Pham, Ngoc Duy; Shang, Jing; Yang, Yang et al.	2020		32	10.1016/j.nanoen.2019.104412
Efficient and stable perovskite solar cells thanks to dual functions of oleyl amine-coated PbSO₄(PbO)₄ quantum dots: Defect passivation and moisture/oxygen blocking	Chen, Chong; Li, Fumin; Zhu, Liangxin et al.	2020		32	10.1016/j.nanoen.2019.104313
Moisture stability in nanostructured perovskite solar cells	Mahdi Malekshahi Byranvand; Ali Nemati Kharat; Nima Taghavinia	2018	Materials Letters	31	10.1016/j.matlet.2018.10.029
Fluorinated Quasi 2D Perovskite Solar Cells with Improved Stability and Over 19% Efficiency	Qiaohui Li; Li Zhou; Tong Zhou	2024	Advanced Energy Materials	31	10.1002/aenm.202400050
Gelation of Hole Transport Layer to Improve the Stability of Perovskite Solar Cells	Ying Zhang; Chenxiao Zhou; Lizhi Lin et al.	2023	Nano-Micro Letters	31	10.1007/s40820-023-01145-y
Improved stability and efficiency of perovskite solar cells with submicron flexible barrier films deposited in air	Nicholas Rolston; Adam D. Printz; F. Hilt et al.	2017		31	10.1039/C7TA09178H
Recent Advances in the Combined Elevated Temperature, Humidity, and Light Stability of Perovskite Solar Cells	Jing Zhou; You Gao; Yongyan Pan et al.	2022	Solar RRL	31	10.1002/solr.202200772
ALD Al₂O₃ on hybrid perovskite solar cells: Unveiling the growth mechanism and long-term stability	Singh, Roja; Ghosh, Sudeshna; Subbiah, Anand S. et al.	2020		31	10.1016/j.solmat.2019.110289
Polarity regulation for stable 2D-perovskite-encapsulated high-efficiency 3D-perovskite solar cells	Su, Hang; Zhang, Lu; Liu, Yucheng et al.	2022		31	10.1016/j.nanoen.2022.106965
Chemical vapor deposition in fabrication of robust and highly efficient perovskite solar cells based on single-walled carbon nanotubes counter electrodes	Van-Dang Tran; S. Pammi; Van-Duong Dao et al.	2018		30	10.1016/J.JALLCOM.2018.02.006
Improving the stability and decreasing the trap state density of mixed-cation perovskite solar cells through compositional engineering	Kianoosh Poorkazem; T. Kelly	2018		30	10.1039/C8SE00127H
Simultaneous Suppression of Multilayer Ion Migration via Molecular Complexation Strategy toward High-Performance Regular Perovskite Solar Cells.	Qian Zhou; Yingying Yang; D. He et al.	2024	Angewandte Chemie	30	10.1002/anie.202416605
Surface Crystallization Modulation toward Highly-Oriented and Phase-Pure 2D Perovskite Solar Cells	Bin Chen; Ke Meng; Zhi-li Qiao et al.	2024	Advances in Materials	30	10.1002/adma.202312054

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Cross-linkable molecule in spatial dimension boosting water-stable and high-efficiency perovskite solar cells	Xi, Jiachen; Wu, Yeyong; Chen, Weijie et al.	2022		30	10.1016/j.nanoen.2021.106846
High-performance quasi-2D perovskite solar cells with power conversion efficiency over 20% fabricated in humidity-controlled ambient air	Lai, Xue; Li, Wenhui; Gu, Xiaoyu et al.	2022		30	10.1016/j.cej.2021.130949
Binary Microcrystal Additives Enabled Antisolvent-Free Perovskite Solar Cells with High Efficiency and Stability	Wang, Deng; Chen, Jiabang; Zhu, Peide et al.	2023		30	10.1002/aenm.202203649
The influence of secondary solvents on the morphology of a spiro-MeOTAD hole transport layer for lead halide perovskite solar cells	L. Ono; Zafer Hawash; E. J. Juárez-Pérez et al.	2018	Journal of Physics D: Applied Physics	29	10.1088/1361-6463/aacb6e
Idealizing Air-Processed Perovskite Film Competitive by Surface Lattice Etching-Reconstruction for High-Efficiency Solar Cells.	Hui Li; Jialong Duan; Chenlong Zhang et al.	2024	Angewandte Chemie	29	10.1002/anie.202419061
High performance planar perovskite solar cells based on CH 3 NH 3 PbI 3-x (SCN) x perovskite film and SnO 2 electron transport layer prepared in ambient air with 70% humility	Yang Zhou; Zongbao Zhang; Yangyang Cai et al.	2018		29	10.1016/J.ELECTACTA.2017.12.076
Self-polymerized Spiro-Type Interfacial Molecule toward Efficient and Stable Perovskite Solar Cells.	Qiushuang Tian; Jingxi Chang; Junbo Wang et al.	2024	Angewandte Chemie	29	10.1002/anie.202318754
Highly stable and Efficient Perovskite Solar Cells Based on FAMA-Perovskite-Cu:NiO Composites with 20.7% Efficiency and 80.5% Fill Factor	Wang, Yousheng; Mahmoudi, Tahmineh; Hahn, Yoon-Bong	2020		29	10.1002/aenm.202000967
Multi-Functional Regulation on Buried Interface for Achieving Efficient Triple-Cation Perovskite Solar Cells.	Yang Ding; X. Feng; Erming Feng et al.	2024	Small	28	10.1002/smll.202308836
Unraveling the Heat- and UV-Induced Degradation of Mixed Halide Perovskite Thin Films via Surface Analysis Techniques	Pei-Chen Huang; Ting-Jia Yang; Chia-Jou Lin et al.	2024	Langmuir	28	10.1021/acs.langmuir.3c03816
Toward Commercialization of Stable Devices: An Overview on Encapsulation of Hybrid Organic-Inorganic Perovskite Solar Cells	Aranda, Clara A.; Caliò, Laura; Salado, Manuel	2021		28	10.3390/cryst11050519
SrTiO₃/Al₂O₃-Graphene Electron Transport Layer for Highly Stable and Efficient Composites-Based Perovskite Solar Cells with 20.6% Efficiency	Mahmoudi, Tahmineh; Wang, Yousheng; Hahn, Yoon-Bong	2020		28	10.1002/aenm.201903369
Efficient and stable CH₃NH₃PbI_{3-x}(SCN)_x planar perovskite solar cells fabricated in ambient air with low-temperature process	Zhang, Zongbao; Zhou, Yang; Cai, Yangyang et al.	2018		28	10.1016/j.jpowsour.2017.11.070
Simultaneously enhanced efficiency and ambient stability of inorganic perovskite solar cells by employing tetramethylammonium chloride additive in CsPbI2Br	I. Jin; Bhaskar Parida; J. Jung	2021	Journal of Materials Science & Technology	27	10.1016/j.jmst.2021.05.084
Unveiling the role of carbon black in printable mesoscopic perovskite solar cells	Priyanka Kajal; Jia Haur Lew; A. Kanwat et al.	2021	Journal of Power Sources	27	10.1016/J.JPOWSOUR.2021.230019
Triple-Additive Strategy for Enhanced Material and Device Stability in Perovskite Solar Cells	Zhenghong Xiong; Yun-Sung Jeon; Hongguang Wang et al.	2025	Advances in Materials	27	10.1002/adma.202413712
Basis and effects of ion migration on photovoltaic performance of perovskite solar cells	Zhou, Wenke; Gu, Juan; Yang, Zhiqian et al.	2021		27	10.1088/1361-6463/abbf74
Efficient and stable perovskite solar cells via surface passivation of an ultrathin hydrophobic organic molecular layer	Li, Jing; Bu, Tongle; Lin, Zhipeng et al.	2021		27	10.1016/j.cej.2020.126712
Enhanced efficiency and thermal stability of perovskite solar cells using poly(9-vinylcarbazole) modified perovskite/PCBM interface	Jiankai Zhang; Wujian Mao; Jiaji Duan et al.	2019	Electrochimica Acta	26	10.1016/J.ELECTACTA.2019.06.102

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NH_4PF_6 assisted buried interface defect passivation for planar perovskite solar cells with efficiency exceeding 21%	Xing-Dong Ding; Xiaowen Zhou; Jinwei Meng et al.	2023	Rare Metals	26	10.1007/s12598-023-02394-x
Efficient and Moisture Resistant Wide-Bandgap Perovskite Solar Cells with Phosphinate-Based Iodine Defect Passivation	Song, Yuting; Liu, Ziyan; Cai, Xinhang et al.	2025		26	10.1002/aenm.202500650
Grain Boundary Passivation with Dion–Jacobson Phase Perovskites for High-Performance Pb–Sn Mixed Narrow-Bandgap Perovskite Solar Cells	Lei Zhang; Qiao Kang; Yanping Song et al.	2021	Solar RRL	25	10.1002/solr.202000681
Sb2S3 and Cu3SbS4 nanocrystals as inorganic hole transporting materials in perovskite solar cells	F. Mohamadkhani; Maryam Heidariramsheh; S. Javadpour et al.	2021	Solar Energy	25	10.1016/J.SOLENER.2021.05.049
5-Chloro-2-Hydroxypyridine Derivatives with Push-Pull Electron Structure Enable Durable and Efficient Perovskite Solar Cells	Can Zheng; Lidan Liu; Yong Li et al.	2023	Advanced Energy Materials	25	10.1002/aenm.202301302
One-Step Construction of a Perovskite/TiO2 Heterojunction toward Highly Stable Inverted All-Layer-Inorganic CsPbI2Br Perovskite Solar Cells with 17.1% Efficiency	Xingyu Pu; Qipeng Cao; Jie Su et al.	2023	Advanced Energy Materials	25	10.1002/aenm.202301607
Evaporated Undoped Spiro-OMeTAD Enables Stable Perovskite Solar Cells Exceeding 20% Efficiency	Du, Guozheng; Yang, Li; Zhang, Cuiping et al.	2022		25	10.1002/aenm.202103966
Two-in-one additive-engineering strategy for improved air stability of planar perovskite solar cells	Yu, Wei; Yu, Shuwen; Zhang, Jing et al.	2018		25	10.1016/j.nanoen.2017.12.041
High-performance fully-ambient air processed perovskite solar cells using solvent additive	Hussein, Haitham T.; Zamel, Rafid S.; Mohamed, Mayada S. et al.	2021		25	10.1016/j.jpcs.2020.109792
Nonstoichiometric Adduct Approach for High-Efficiency Perovskite Solar Cells.	N. Park	2017	Inorganic Chemistry	24	10.1021/acs.inorgchem.6b01294
Degradation of inverted architecture CH3NH3PbI3-xClx perovskite solar cells due to trapped moisture	Christopher Bracher; B. Freestone; D. Mohamad et al.	2018		24	10.1002/ese3.180
Fast Fabrication of a Stable Perovskite Solar Cell with an Ultrathin Effective Novel Inorganic Hole Transport Layer.	Aibin Huang; Lei Lei; Jingting Zhu et al.	2017	Langmuir	24	10.1021/acs.langmuir.7b00127
Self-assembled interlayer aiming at the stability of NiO_x based perovskite solar cells	Guo, Tonghui; Fang, Zhi; Zhang, Zequn et al.	2022		24	10.1016/j.jechem.2022.01.049
Encapsulation of Perovskite Solar Cells for High Humidity Conditions	Qi Dong; Fangzhou Liu; Man Kwong Wong et al.	2016	ChemSusChem	23	10.1002/cssc.201601090
Formamidinium-incorporated Dion-Jacobson phase 2D perovskites for highly efficient and stable photovoltaics	Sajjad Ahmad; Sajjad Ahmad; Wei Yu et al.	2021	Journal of Energy Chemistry	23	10.1016/j.jechem.2020.08.055
All-inorganic, hole-transporting-layer-free, carbon-based CsPbIBr2 planar perovskite solar cells by a two-step temperature-control annealing process	Cheng Wang; Jun-Sen Zhang; J. Duan et al.	2020		23	10.1016/j.mssp.2019.104870
Stability of organometal halide perovskite solar cells and role of HTMs: recent developments and future directions	Ehsan Raza; F. Aziz; Z. Ahmad	2018	RSC Advances	23	10.1039/c8ra03477j
A novel perylene diimide-based zwitterion as the cathode interlayer for high-performance perovskite solar cells	Helin Wang; Jun Song; J. Qu et al.	2020		23	10.1039/d0ta06006b
Development of hermetic glass frit encapsulation for perovskite solar cells	Emami, Seyedali; Martins, Jorge; Madureira, Ruben et al.	2019		23	10.1088/1361-6463/aaf1f4
A pressure process for efficient and stable perovskite solar cells	Luo, Junsheng; Xia, Jianxing; Yang, Hua et al.	2020		23	10.1016/j.nanoen.2020.105063

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Highly efficient and stable inorganic CsPbBr₃ perovskite solar cells via vacuum co-evaporation	Duan, Yanyan; Zhao, Gen; Liu, Xiaotao et al.	2021		23	10.1016/j.apsusc.2021.150153
Improving performance and moisture stability of perovskite solar cells through interface engineering with polymer-2D MoS₂ nanohybrid	Ray, Rajeev; Sarkar, Abdus Salam; Pal, Suman Kalyan	2019		23	10.1016/j.solener.2019.09.055
The stable perovskite solar cell prepared by rapidly annealing perovskite film with water additive in ambient air	Tingwei He; Zhiyong Liu; Yu Zhou et al.	2018		22	10.1016/J.SOLMAT.2017.12.015
Fast Charge Transfer and High Stability via Hybridization of Hygroscopic Cu-BTC Metal-Organic Framework Nanocrystals with a Light-Absorbing Layer for Perovskite Solar Cells.	Jiyoung Lee; Nikolai Tsvetkov; S. Shin et al.	2022	ACS Applied Materials and Interfaces	22	10.1021/acsami.2c05488
Accelerated Degradation of FAPbI3 Perovskite by Excess Charge Carriers and Humidity	Seo-Ryeong Lee; Donghyeon Lee; Seung-Gu Choi et al.	2024	Solar RRL	22	10.1002/solr.202300958
Ambient condition-processing strategy for improved air-stability and efficiency in mixed-cation perovskite solar cells	I. Asuo; D. Gedamu; N. Y. Doumon et al.	2020		22	10.1039/d0ma00528b
Cyclic Multi-Site Chelation for Efficient and Stable Inverted Perovskite Solar Cells.	Jiandong He; Shuai Yang; Chao Luo et al.	2024	Angewandte Chemie	22	10.1002/anie.202414118
Transition Metal-Oxide Free Perovskite Solar Cells Enabled by a New Organic Charge Transport Layer.	Sehoon Chang; Ggoch Ddeul Han; J. Weis et al.	2016	ACS Applied Materials and Interfaces	22	10.1021/acsami.6b00635
Polymer-modified CsPbI2Br films for all-inorganic planar perovskite solar cells with improved performance	Caiyun Zhang; Xiaojin Wan; J. Zang et al.	2021		22	10.1016/j.surfin.2020.100809
Double-layer synergistic optimization by functional black phosphorus quantum dots for high-efficiency and stable planar perovskite solar cells	Zhang, Yuhong; Xu, Lin; Wu, Yanjie et al.	2021		22	10.1016/j.nanoen.2021.106610
Enhancing the stability of planar perovskite solar cells by green and inexpensive cellulose acetate butyrate	Xiao, Bo; Qian, Yongxin; Li, Xin et al.	2023		22	10.1016/j.jechem.2022.09.039
Research Progress on Stability of FAPbI3 Perovskite Solar Cells	Wenxin Deng; Jianwei Wei; Zengwei Ma et al.	2025	Crystal research and technology (1981)	21	10.1002/crat.202400228
Full-scale chemical and field-effect passivation: 21.52% efficiency of stable MAPbI3 solar cells via benzenamine modification	Fengyou Wang; Meifang Yang; Yuhong Zhang et al.	2021	Nano Reseach	21	10.1007/s12274-021-3286-2
Recent strategies to improve moisture stability in metal halide perovskites materials and devices	Zhou, Chenxiao; Tarasov, Alexey B.; Goodilin, Eugene A. et al.	2022		21	10.1016/j.jechem.2021.05.035
Understanding the thermal degradation mechanism of perovskite solar cells via dielectric and noise measurements	Kumar, Ankit; Bansode, Umesh; Ogale, Satishchandra et al.	2020		21	10.1088/1361-6528/ab97d4
Indoloindole-Based Hole Transporting Material for Efficient and Stable Perovskite Solar Cells Exceeding 24% Power Conversion Efficiency	Kim, Dong Won; Choi, Kang-Hoon; Hong, Seung Hwa et al.	2023		21	10.1002/aenm.202300219
Multifunctional 2D perovskite capping layer using cyclohexylmethylammonium bromide for highly efficient and stable perovskite solar cells	Wang, Shibo; Cao, Fengxian; Wu, Yunjia et al.	2021		21	10.1016/j.mtphys.2021.100543
A Porphyrin-Involved Benzene-1,3,5-Tricarboxamide Dendrimer (Por-BTA) as a Multifunctional Interface Material for Efficient and Stable Perovskite Solar Cells.	Kuo-Pin Su; Penghui Zhao; Yu Ren et al.	2021	ACS Applied Materials and Interfaces	20	10.1021/acsami.1c00146
Progress towards highly stable and lead-free perovskite solar cells	Abd Mutalib, Muhazri; Ahmad Ludin, Norasikin; Nik Ruzalman, Nik Ahmad Aizudden et al.	2018		20	10.1007/s40243-018-0113-0

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Reducing energy loss and stabilising the perovskite/poly (3-hexylthiophene) interface through a polyelectrolyte interlayer	Wenxiao Zhang; L. Wan; Sheng Fu et al.	2020		19	10.1039/d0ta01860k
A Thiophene Based Dopant-Free Hole-Transport Polymer for Efficient and Stable Perovskite Solar Cells	In-Bok Kim; Yeon-Ju Kim; Dong-Yu Kim et al.	2022	Macromolecular Research	19	10.1007/s13233-022-0042-8
Regulating Spatial Charge Distribution of Multisite Passivating Additives Enables Stable Perovskite Solar Cells	Xinyu Tong; Yujiao Ma; Lisha Xie et al.	2025	Advanced Functional Materials	19	10.1002/adfm.202511458
Analysis of degradation kinetics of halide perovskite solar cells induced by light and heat stress	Khadka, Dhruba B.; Shirai, Yasuhiro; Yanagida, Masatoshi et al.	2022		19	10.1016/j.solmat.2022.111899
Perfluoroalkylsulfonyl ammonium for humidity- resistant printing high-performance phase-pure FAPbI₃ perovskite solar cells and modules	Chen, Xining; Yang, Fu; Yuan, Linhao et al.	2024		19	10.1016/j.joule.2024.05.018
Improving the stability of metal halide perovskite solar cells from material to structure	Qin, Kui; Dong, Binghai; Wang, Shimin	2019		19	10.1016/j.jechem.2018.08.004
In situ growth of a 2D/3D mixed perovskite interface layer by seed-mediated and solvent-assisted Ostwald ripening for stable and efficient photovoltaics	Shiqi Li; Lina Hu; Chenxi Zhang et al.	2020		18	10.1039/c9tc05212g
Merged interface construction toward ultra-low Voc loss in inverted two-dimensional Dion–Jacobson perovskite solar cells with efficiency over 18%	Haotian Wu; Xiaomei Lian; Jun Li et al.	2021		18	10.1039/D1TA02015C
Band Tailing and Deep Defect States in CH 3 NH 3 Pb (I 1 − x Br x) 3 Perovskites As Revealed by Sub-Bandgap Photocurrent	C. Sutter-Fella; D. W. Miller; Quynh P. Ngo et al.	2017		18	(link)
Universal Strategy with Structural and Chemical Crosslinking Interface for Efficient and Stable Perovskite Solar Cells	Keqing Huang; Li-chun Chang; Yihui Hou et al.	2024	Advanced Energy Materials	18	10.1002/aenm.202304073
Superiority of two-step deposition over one-step deposition for perovskite solar cells processed in high humidity atmosphere	M. F. Mohamad Noh; Nurul Affiqah Arzaee; Inzamam Nawas Nawas Mumthas et al.	2021	Optical materials (Amsterdam)	18	10.1016/j.optmat.2021.111288
High efficient and long-time stable planar heterojunction perovskite solar cells with doctor-bladed carbon electrode	Man Yang; Jun Yu Li; Jinhua Li et al.	2019	Journal of Power Sources	17	10.1016/J.JPOWSOUR.2019.03.092
Simulation and Optimization of Lead-Based Perovskite Solar Cells with Cuprous Oxide as a P-type Inorganic Layer	D. Eli; M. Onimisi; S. Garba et al.	2019	Journal of the Nigerian Society of Physical Sciences	17	10.46481/jnsps.2019.13
Dopant-free polymer/2D/3D perovskite solar cells with high stability	Xiaoqing Jiang; Jiafeng Zhang; Yang Liu et al.	2021	Nano Energy	17	10.1016/j.nanoen.2021.106521
Green solvent engineering for enhanced performance and reproducibility in printed carbon-based mesoscopic perovskite solar cells and modules	C. Worsley	2022	Proceedings of the International Conference on Hybrid and Organic Photovoltaics	17	10.29363/nanoge.hopv.2022.038
22.1% Carbon-Electrode Perovskite Solar Cells by Spontaneous Passivation and Self-Assembly of Hole-Transport Bilayer.	Junjie Tong; Chen Dong; Miaosen Yao et al.	2025	ACS Nano	17	10.1021/acsnano.4c16916
Making fully printed perovskite solar cells stable outdoor with inorganic superhydrophobic coating	Luo, Jianqiang; Yang, Hong Bin; Zhuang, Mingxiang et al.	2020		17	10.1016/j.jechem.2020.03.082
Intercepting the Chelation of Perovskites with Ambient Moisture through Active Addition Reaction for Full-Air-Processed Perovskite Solar Cells	Ning, Lei; Song, Lixin; Yao, Zhengzheng et al.	2024		17	10.1002/aenm.202401320
Additive engineering enabled non-radiative defect passivation with improved moisture-resistance in efficient and stable perovskite solar cells	Azam, Muhammad; Ke, Zhicheng; Luo, Junsheng et al.	2024		17	10.1016/j.cej.2024.149424

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Impact of Electrode Materials on Process Environmental Stability of Efficient Perovskite Solar Cells	Chung, Jaehoon; Shin, Seong Sik; Kim, Geunjin et al.	2019		17	10.1016/j.joule.2019.05.018
Unveiling the irreversible performance degradation of organo-inorganic halide perovskite films and solar cells during heating and cooling processes.	A. Mamun; T. T. Ava; H. Byun et al.	2017	Physical Chemistry, Chemical Physics - PCCP	16	10.1039/c7cp03106h
Boosting Photovoltaic Efficiency: The Role of Functional Group Distribution in Perovskite Film Passivation.	Qingquan He; Shicheng Pan; Tao Zhang et al.	2024	Small	16	10.1002/smll.202410481
Hydrophobic 4-(isopropylbenzyl)oxy-substituted metallophthalocyanines as a dopant-free hole selective material for high-performance and moisture-stable perovskite solar cells	Kong, Fantai; Güzel, Emre; Sonmezoglu, Savas	2023		16	10.1016/j.mtener.2023.101324
Vertical distribution of PbI₂ nanosheets for robust air-processed perovskite solar cells	Zhu, Zhenkun; Shang, Jitao; Tang, Guanqi et al.	2023		16	10.1016/j.cej.2022.140163
Multidentate anchoring strategy for synergistically modulating crystallization and stability towards efficient perovskite solar cells	Hu, Ping; Zhou, Wenbo; Chen, Junliang et al.	2024		16	10.1016/j.cej.2023.148249
Li-TFSI endohedral Metal-Organic frameworks in stable perovskite solar cells for Anti-Deliquescent and restricting ion migration	Wang, Jiaqi; Zhang, Jian; Yang, Yulin et al.	2022		16	10.1016/j.cej.2021.132481
Humidity-insensitive fabrication of efficient perovskite solar cells in ambient air	Wang, Feng; Ye, Zongbiao; Sarvari, Hojjatollah et al.	2019		16	10.1016/j.jpowsour.2018.11.013
Healing soft interface for stable and high-efficiency all-inorganic CsPbIBr₂ perovskite solar cells enabled by S-benzylisothiourea hydrochloride	Yan, Furi; Yang, Peizhi; Li, Jiabao et al.	2022		16	10.1016/j.cej.2021.132781
Modulation of Perovskite Grain Boundaries by Electron Donor–Acceptor Zwitterions R,R-Diphenylamino-phenyl-pyridinium-(CH2)n-sulfonates: All-Round Improvement on the Solar Cell Performance	C. Hung; Jin-Tai Lin; Yu-Hsuan Yang et al.	2022	JACS Au	15	10.1021/jacsau.2c00160
CsSnI3 Quantum Dots as a Multifunctional Interlayer for High-Efficiency Bilayer Perovskite Solar Cells	Chunchen Liu; Xiaojian Zhen; Wenyuan Peng et al.	2025	Advanced Energy Materials	15	10.1002/aenm.202405074
Bilayer interfacial engineering with PEAI/OAI for synergistic defect passivation in high-performance perovskite solar cells	Chentai Cao; Yuli Tao; Quan Yang et al.	2025	Journal of Semiconductors	15	10.1088/1674-4926/25030046
Enhancing efficiency of perovskite solar cells from surface passivation of Co2+ doped CuGaO2 nanocrystals.	Ruoshui Li; Yanfei Dou; Yinsheng Liao et al.	2021	Journal of Colloid and Interface Science	15	10.1016/j.jcis.2021.09.102
Ionic liquid doped organic hole transporting material for efficient and stable perovskite solar cells	M. Elawad; Husileng Lee; Ze Yu et al.	2020		15	10.1016/j.physb.2020.412124
In-situ observation of moisture-induced degradation of perovskite solar cells using laser-beam induced current	Zhaoning Song; A. Abate; Suneth C. Watthage et al.	2016	Photovoltaic Specialists Conference	15	10.1109/PVSC.2016.7749805
Roll-transferred graphene encapsulant for robust perovskite solar cells	Yi, Ahra; Chae, Sangmin; Won, Sejeong et al.	2020		15	10.1016/j.nanoen.2020.105182
MAPbI₃/agarose photoactive composite for highly stable unencapsulated perovskite solar cells in humid environment	Yang, Ying; Chen, Tian; Pan, Dequn et al.	2020		15	10.1016/j.nanoen.2019.104246
CoFe2O4 nanocrystals for interface engineering to enhance performance of perovskite solar cells	R. Li; Yinsheng Liao; Yanfei Dou et al.	2021		14	10.1016/J.SOLENER.2021.03.073

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Stabilization Techniques of Lead Halide Perovskite for Photovoltaic Applications	Y. Zheng; Shuang Yang	2021	Solar RRL	14	10.1002/solr.202100710
Implementation of a Multi-Functional-Group Strategy for Enhanced Performance of Perovskite Solar Cells through the Incorporation of 3-Amino-4-Phenylbutyric Acid Hydrochloride.	Danyang Qi; Yang Cao; Xiaolong Feng et al.	2024	Small	14	10.1002/smll.202401487
Emerging Halide Perovskite Materials and Devices for Optoelectronics	Tae-Woo Lee	2019	Advances in Materials	14	10.1002/adma.201905077
A highly stable hole-conductor-free Cs MA1-PbI3 perovskite solar cell based on carbon counter electrode	Shaowei Wang; Huijing Liu; H. Bala et al.	2020		14	10.1016/j.electacta.2020.135686
Charge transfer enhancement of TiO2/perovskite interface in perovskite solar cells	Xuewen Sun; Meng Li; Qinyuan Qiu et al.	2021	Journal of Materials Science: Materials in Electronics	14	10.1007/s10854-021-06778-6
Designing conductive fullerenes ionene polymers as efficient cathode interlayer to improve inverted perovskite solar cells efficiency and stability	Tian Zheng; Hang Zhou; Bin Fan et al.	2021	Chemical Engineering Journal	14	10.1016/J.CEJ.2021.128816
Scalable synthesis of Ti-doped MoO₂ nanoparticle-hole-transporting-material with high moisture stability for CH₃NH₃PbI₃ perovskite solar cells	Im, Kyungmin; Heo, Jin Hyuck; Im, Sang Hyuk et al.	2017		14	10.1016/j.cej.2017.07.160
Improvement of the stability of perovskite solar cells in terms of humidity/heat via compositional engineering	Valipour, F.; Yazdi, E.; Torabi, N. et al.	2020		14	10.1088/1361-6463/ab8511
Anhydrous organic etching derived fluorine-rich terminated MXene nanosheets for efficient and stable perovskite solar cells	Zhao, Yu; Li, Bin; Tian, Chuanming et al.	2023		14	10.1016/j.cej.2023.143862
Interface Engineering with Formamidinium Salts for Improving Ambient-Processed Inverted CsPbI3 Photovoltaic Performance: Intermediate- vs Post-Treatment.	Tianxiang Li; Wan Li; Kun Wang et al.	2023	ACS Applied Materials and Interfaces	13	10.1021/acsami.3c10768
Resist Thermal Shock Through Viscoelastic Interface Encapsulation in Perovskite Solar Cells	Sai Ma; Jiahong Tang; Guizhou Yuan et al.	2024	ENERGY & ENVIRONMENTAL MATERIALS	13	10.1002/eem2.12739
Limitations and Progresses in Carbon-Based Cesium Lead Halide Perovskite Solar Cells.	Wenran Wang; Jianxin Zhang; Huishi Guo et al.	2024	ChemSusChem	13	10.1002/cssc.202301761
Self-Formed Multifunctional Grain Boundary Passivation Layer Achieving 22.4% Efficient and Stable Perovskite Solar Cells	Wenqi Wang; Qian Zhou; D. He et al.	2021	Solar RRL	13	10.1002/solr.202100893
Degradation mechanism and stability improvement of formamidine-based perovskite solar cells under high humidity conditions	Cao, Fengren; Zhang, Peng; Sun, Haoxuan et al.	2022		13	10.1007/s12274-022-4524-y
Efficient and stable perovskite solar cells through e-beam preparation of cerium doped TiO₂ electron transport layer, ultraviolet conversion layer CsPbBr₃ and the encapsulation layer Al₂O₃	Jin, Junjie; Li, Hao; Bi, Wenbo et al.	2020		13	10.1016/j.solener.2020.01.048
Highly stable perovskite solar cells with a novel Ni-based metal organic complex as dopant-free hole-transporting material	Tai Wu; Linqin Wang; Rongjun Zhao et al.	2022		12	10.1016/J.JECHEM.2021.06.005
-Conjugated Polymer with Pendant Side Chains as a Dopant-Free Hole Transport Material for High-Performance Perovskite Solar Cells.	Zhiqing Xie; Jeonghyeon Park; Hyerin Kim et al.	2024	ACS Applied Materials and Interfaces	12	10.1021/acsami.3c15611

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A highly hydrophobic fluorographene-based system as an interlayer for electron transport in organic-inorganic perovskite solar cells	Saqib Javaid; C. Myung; Saeed Pourasad et al.	2018		12	10.1039/C8TA05811C
Impact of halide additives on green antisolvent and high-humidity processed perovskite solar cells	Qian Chen; J. C. Ke; D. Wang et al.	2021		12	10.1016/J.APSUSC.2020.147949
Environmental risks and strategies for the long-term stability of carbon-based perovskite solar cells	Meng, F.; Zhou, Y.; Gao, L. et al.	2021		12	10.1016/j.mtener.2020.100590
Efficient and stable perovskite solar cells: The effect of octadecyl ammonium compound side-chain	Feng, Zhiying; Weng, Chaocang; Hua, Yikun et al.	2023		12	10.1016/j.cej.2023.146498
Enhanced anchoring enables highly efficient and stable inverted perovskite solar cells	Yin, Ran; Wu, Rongfei; Miao, Wenjing et al.	2024		12	10.1016/j.nanoen.2024.109544
Morphology control of perovskite film for efficient CsPbIBr₂ based inorganic perovskite solar cells	Pan, Junye; Zhang, Xing; Zheng, Yong et al.	2021		12	10.1016/j.solmat.2020.110878
Highly stable and efficient planar perovskite solar cells using ternary metal oxide electron transport layers	Thambidurai, M.; Shini, Foo; Harikesh, P. C. et al.	2020		12	10.1016/j.jpowsour.2019.227362
Biomass-derived functional additive for highly efficient and stable lead halide perovskite solar cells with built-in lead immobilisation	Jing Li; Xiang Qiao; Bingchen He et al.	2025	Energy & Environmental Science	11	10.1039/d4ee06038e
Efficient and Stable Perovskite Large Area Cells by Low-Cost Fluorene-Xantene-Based Hole Transporting Layer	L. Vesce; M. Stefanelli; A. Di Carlo	2021	Energies	11	10.3390/en14196081
Efficiency and Stability Enhancement of Fully Ambient Air Processed Perovskite Solar Cells Using TiO2 Paste with Tunable Pore Structure	Seyedeh Mozhgan Seyed-Talebi; I. Kazeminezhad; Saeed Shahbazi et al.	2019	Advanced Materials Interfaces	11	10.1002/admi.201900939
Optimization of Cs2SnBr6/Cs2TeI6 bilayer absorber for enhanced performance in perovskite solar cell	Manasvi Raj; Anshul Aggarwal; Aditya Kushwaha et al.	2025	Physica Scripta	11	10.1088/1402-4896/ae1b44
Enhancing the hydrophobicity of perovskite solar cells using C18 capped CH3NH3PbI3 nanocrystals		2018		11	10.1039/C8TC01939H
Enhanced Light Absorption by Facile Patterning of Nano-Grating on Mesoporous TiO2 Photoelectrode for Cesium Lead Halide Perovskite Solar Cells	Kang-Pil Kim; W. Kim; S. Kwon et al.	2020	Nanomaterials	11	10.3390/nano11051233
Fabrication of Efficient and Stable Perovskite Solar Cells in High-Humidity Environment through Trace-Doping of Large-Sized Cations	Liu, Xueping; He, Jiang; Wang, Pengfei et al.	2019		11	10.1002/cssc.201900106
Degradation mechanisms and stability challenges in perovskite solar cells: a comprehensive review	Sharshir, Swellam W.; El-Naggar, Ahmed A.; Ismail, Hamdy A. et al.	2025		11	10.1016/j.solener.2025.113707
Hydrophobic hole-transporting layer induced porous PbI₂ film for stable and efficient perovskite solar cells in 50% humidity	Hou, Fuhua; Jin, Fangming; Chu, Bei et al.	2016		11	10.1016/j.solmat.2016.08.024
Multifunctional Interface Modification Enables Efficient and Stable HTL-Free Carbon-Electroded CsPbI2Br Perovskite Solar Cells.	Wei Zhao; Lin Wu; Jianlin Chen et al.	2024	ChemSusChem	10	10.1002/cssc.202400223
Halogenated Chiral Organic Spacer Cation Regulation for Efficient and Stable 2D Ruddlesden-Popper Perovskite Solar Cells	Xin Li; Xue Dong; Zihong Shen et al.	2025	Advanced Functional Materials	10	10.1002/adfm.202507591

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Two-step processed efficient perovskite solar cells via improving perovskite/PTAA interface using solvent engineering in PbI2 precursor	Caoyu Long; Ning Wang; Keqing Huang et al.	2020		10	10.1088/1674-1056/ab7744
A Chelating-Agent-Passivated Electron Transport Layer for Efficient Perovskite Solar Cells with Enhanced Reproducibility	Shui Wang; Runying Dai; Xiangchuan Meng et al.	2023	Advanced Functional Materials	10	10.1002/adfm.202310860
Performance enhancement of CsPbI2Br perovskite solar cells via stoichiometric control and interface engineering	Kalsoom Fatima; Muhammad Irfan Haider; A. Fakharuddin et al.	2020		10	10.1016/j.solener.2020.10.007
Importance and Advancement of Modification Engineering in Perovskite Solar Cells	Xiang Wu; Bo Wu; Zhiguo Zhu et al.	2022	Solar RRL	10	10.1002/solr.202200171
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Multifunctional sodium phytate as buried interface Passivator for high efficiency and stable planar perovskite solar cells	Su, Haijun; Liu, Congcong; Fan, Huichao et al.	2024		10	10.1016/j.cej.2024.157212
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Sustainable Self-Healing of Perovskite Solar Cells Using Dendrimers as Volatile Reservoirs	Bonkee Koo; Wooyeon Kim; Kiwon Choi et al.	2025	Advances in Materials	9	10.1002/adma.202512410
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<inline-formula><mml:math><mml:mi>Si</mml:mi></mml:math></inline-formula>/diamond thermal boundary conductance enhanced by <inline-formula><mml:math><mml:msub><mml:mrow><mml:mi>Si</mml:mi><mml:mi>N</mml:mi></mml:mrow><mml:mi>x</mml:mi></mml:ms formula> and amorphous carbon interlayers	Adnan, Khalid Zobaid; Abedin, Tanvir; Feng, Tianli	2026		1	10.1103/84jf-svtg
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Subdiffusion of nonlinear waves in quasiperiodic potentials	Marco Larcher; Tetyana V. Laptyeva; Joshua D. Bodyfelt et al.	2012	New Journal of Physics 14 (2012) 103036	0	10.1088/1367-2630/14/10/103036
Diffraction and pseudospectra in non-Hermitian quasiperiodic lattices	Ananya Ghatak; Dimitrios H. Kaltsas; Manas Kulkarni et al.	2024	arXiv preprint	0	(link)

Title	Authors	Year	Venue	Cited	DOI
Discrete quasiperiodic sets with predefined covering cluster	Nicolae Cotfas	2005	Philosophical Magazine 86 (2006) 895-900	0	10.1080/14786430500263413
Localization control born of intertwined quasiperiodicity and non-Hermiticity	Jeon, Junmo; Lee, SungBin	2022	SciPost Phys.Core	0	10.21468/SciPostPhysCore.6.4.077
Topological Pumping over a Photonic Fibonacci Quasicrystal	Verbin, Mor; Zilberberg, Oded; Lahini, Yoav et al.	2014	Phys.Rev.B	0	10.1103/PhysRevB.91.064201
Anderson localization in Bose-Einstein condensates	Modugno, Giovanni	2010	Rept.Prog.Phys.	0	10.1088/0034-4885/73/10/102401
Anomalous localization in a kicked quasicrystal	Shimasaki, Toshihiko; Prichard, Max; Kondakci, H. Esat et al.	2022	Nature Phys.	0	10.1038/s41567-023-02329-4
Support Vector Machine Kernels as Quantum Propagators	Kuo, Nan-Hong; Wong, Renata	2025	INSPIRE record	0	(link)
Colloquium: Synthetic quantum matter in nonstandard geometries	Grass, Tobias; Bercioux, Dario; Bhattacharya, Utso et al.	2024	Rev.Mod.Phys.	0	10.1103/RevModPhys.97.011001
Phasonic Spectroscopy of a Quantum Gas in a Quasicrystalline Lattice	Rajagopal, Shankari V.; Shimasaki, Toshihiko; Dotti, Peter et al.	2019	Phys.Rev.Lett.	0	10.1103/PhysRevLett.123.223201
Non-Hermitian physics	Ashida, Yuto; Gong, Zongping; Ueda, Masahito	2020	Adv.Phys.	0	10.1080/00018732.2021.1876991
Cavity QED with Quantum Gases: New Paradigms in Many-Body Physics	Mivehvar, Farokh; Piazza, Francesco; Donner, Tobias et al.	2021	Adv.Phys.	0	10.1080/00018732.2021.1969727
An exactly solvable model for two-dimensional non-Hermitian quasicrystals	S. Kou	2021	Science China Physics Mechanics and Astronomy	0	10.1007/s11433-021-1824-1
Mie-Resonant Nanophotonics and Metasurfaces	N. Lassaline; Raphael Brechbühler; Deepankur Thureja et al.	2022		0	(link)
Time-Domain Interferometric Measurement of Photonic Band Structure in a One-Dimensional Quasicrystal	T. Hattori; N. Tsurumachi; S. Kawato et al.	1994	Ultrafast Phenomena	0	10.1007/978-3-642-85176-6_103
cc sd-0 00 04 32 5 , v er si on 3-2 2 A ug 2 00 5 Bose-Einstein Condensates in Optical Quasicrystal Lattices	L. Sanchez-Palencia; L. Santos	2005		0	(link)
PRO O F CO PY 076503 APL Photonic quasicrystal single-cell cavity mode	Sun-Kyung Kim; Jeehye Lee; Se-Heon Kim et al.	2004		0	(link)
Program of Oral Presentations	M. Caudle; A. Kreyssig; S. Nandi et al.	2010		0	(link)
Photonic Time Quasicrystals	M. Coppolaro; M. Moccia; G. Castaldi et al.	2025	Metamaterials	0	10.1109/Metamaterials65622.2025.1117
Tuning Capability of THz Surface Modes in Photonic Quasicrystal Containing Graphene	A. Namdar; Rana Feizollahi Onsoroudi; H. Khoshsima et al.	2017		0	(link)
Title Localized modes in photonic quasicrystals with penrose-type lattice Permalink	A. D. Villa; S. Enoch; G. Tayeb et al.	2006		0	(link)
Photonic band gap effect and localization in two-dimensional penrose lattice-Lasers and Electro-Optics, 2001. CLEO '01. Technical Digest. Summaries of papers presented at the	M. Bayindir	2004		0	(link)
Localization of Light in Photonics Lattices for All-Optical Representation of Binaries	Nguyen, Dan T.; Nolan, Daniel A.; Borrelli, Nicholas F.	2021		0	10.3389/fphy.2021.709428
Observation of localization of light in photonic quasicrystals of diverse symmetries	Wang, Peng; Fu, Qidong; Konotop, Vladimir V. et al.	2024		0	10.48550/arXiv.2412.18253
Probing localization in the 2D Aubry-Andre model using ultracold atoms in an optical lattice	Wu, Qijun; Reeve, Lee; Gröters, David et al.	2024		0	(link)
Aperiodic lattices for tunable photonic bandgaps and localization	Chakraborty, Subhasish; Parker, Michael C.; Mears, Robert J.	2005		0	10.48550/arXiv.physics/0507081
Signature of simultaneous onset of topological edge-state and transverse localized state in a 1-D specialty photonic lattice	Bhattacherjee, Sayan; Ghosh, Somnath	2022		0	10.1016/j.optcom.2022.128500
Coherent control of localization in modulated quasiperiodic lattices	Shimasaki, Toshihiko; Kondakci, Hasan; Prichard, Max et al.	2022		0	(link)
Postdoctoral Fellowship: MPS-Ascend: Unveiling Transport and Critical States in Quasicrystalline Optical Lattices and Amplifying Science Engagement Through Digital Storytelling	Wilson, Cedric C	2024		0	(link)

Title	Authors	Year	Venue	Cited	DOI
Effects of Localized Defects / Sources in a Periodic Lattice Using Green's Function of Periodic Scatterers	Tan, Shurun; Tsang, Leung	2018		0	10.1109/APUSNCURSINRSM.2018.860
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Localization and reentrant delocalization transitions in one-dimensional photonic quasicrystals	Vaidya, Sachin; Jörg, Christina; Linn, Kyle et al.	2024		0	(link)
Emergent flat bands and nonlinear phenomena of Bose—Einstein condensates in two-dimensional trilayer moiré optical lattices	Liu, Xiuye; Zeng, Jianhua	2026		0	10.15302/frontphys.2026.042201
A comprehensive Fourier (k-) space design approach for controllable single and multiple photon localization states	Chakraborty1, Subhasish; Parker2, Michael C.; Mears3, Robert J. et al.	2005		0	10.48550/arXiv.physics/0501163
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Phasonic spectroscopy in a tunable quasicrystalline optical lattice	Rajagopal, Shankari; Shimasaki, Toshihiko; Dotti, Peter et al.	2019		0	(link)
Aperiodic lattices in silicon nano-wire for spectrally engineered DWDM photonics	Chakraborty, Subhasish; Parker, M. C.; Irvine, A. C. et al.	2006		0	10.1109/CLEO.2006.4628626
Localised defects in one-dimensional periodic structures and relative phase measurements of the radiation passed through the periodic lattice	Konoplev, I. V.; MacInnes, P.; Cross, A. W. et al.	2008		0	10.1109/PLASMA.2008.4591174
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Driving versus disorder: interplay of dynamic localization and Aubry-André localization in a cold atom quasicrystal	Shimasaki, Toshihiko; Dotti, Peter; Bai, Yifei et al.	2024		0	(link)
Fibonacci Optical Lattices	Singh, Kevin; Geiger, Zachary; Senaratne, Ruwan et al.	2015		0	(link)
Novel 1D photonic crystal biosensor for the detection of tuberculosis in blood samples	Mukherjee, Trinath; Ansari, Mohammad Abu Saqib Alam; Sharma, Anup Kumar et al.	2026		0	10.1007/s11082-026-08904-2
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Experimental Realisation of PT-Symmetric Flat Bands	Biesenthal, Tobias; Kremer, Mark; Heinrich, Matthias et al.	2019		0	10.1109/CLEOE-EQEC.2019.8873246
Localization in a Quasi-Periodic One Dimensional System	Biddle, John; Priour, Donald; Das Sarma, Sankar	2009		0	(link)
Ultracold bosons in 2D quasicrystalline and Aubry-Andre systems	Reeve, Lee; Wu, Qijun; Gröters, David et al.	2024		0	(link)
Eigenstates of Photonic Crystal Structures Visualized in Real Space and in k-Space	Engelen, R. J. P.; Sugimoto, Y.; Gersen, H. et al.	2007		0	10.1109/ICTON.2007.4296164
Coherent control of localization in driven quasicrystals and integer quantum Hall systems	Bai, Yifei; Dotti, Peter; Shimasaki, Toshihiko et al.	2024		0	(link)
Symmetry-induced quasicrystalline waveguides	Davies, Bryn; Craster, Richard V	2022		0	10.48550/arXiv.2208.14906
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Delocalization of a disordered bosonic system by repulsive interactions	Deissler, Benjamin; Zaccanti, Matteo; Roati, Giacomo et al.	2010		0	(link)
Watching pythagorean numbers: Localization-delocalization transition in two-dimensional aperiodic potentials	Huang, Changming; Ye, Fangwei; Chen, Xianfeng et al.	2016		0	10.1109/PIERS.2016.7735253
Anomalous localization in a kicked quasicrystal	Shimasaki, Toshihiko; Prichard, Max; Kondakci, Hasan et al.	2023		0	(link)
Revealing Superfluid–Mott-Insulator Transition in an Optical Lattice	Prokof'ev, Nikolay	2003		0	(link)
Optical quasi crystals-properties and dynamics	Freedman, B.; Bartal, G.; Segev, M. et al.	2005		0	10.1109/QELS.2005.1548893
Light propagation in disordered aperiodic Mathieu photonic lattices	Vasiljević, Jadranka M.; Timotijević, Dejan V.; Jović Savić, Dragana M.	2022		0	10.1051/epjconf/202226608015

Title	Authors	Year	Venue	Cited	DOI
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Acquisition of Instrumentation for the Study of Nonlinear Nanoscale Localization and for Student Training	Sievers, Albert J	2000		0	(link)
Disorder-induced enhancement of nonclassical correlations in programmable integrated quantum walks	Zhang, Zhi-Yuan; Chen, Yang; Feng, Lan-Tian et al.	2026		0	10.1117/1.APN.5.1.016019
Collision dynamics of molecules and rotational excitons in an ultracold gas confined by an optical lattice	Krems, Roman	2010		0	(link)
Transmission properties of a Fibonacci quasi-crystals containing single-negative materials and their usage as multi-channel filters	Charkhesht, Ali; Pashaei Adl, Hamid; Roshan Entezar, Samad	2014		0	(link)
1. LOCALIZATION AND QUANTUM CHAOS: Chaotic quantum transport in superlattices	Fromhold, T. M.; Krokhin, A. A.; Wilkinson, P. B. et al.	2001		0	10.1070/1063-7869/44/10S/S03
Nanoimprinted deterministic aperiodic nanostructures for tailored light matter interactions	Xavier, Jolly; Probst, Jurgen; Becker, Christiane	2017		0	10.1109/CLEOE-EQEC.2017.8087096
Hypermoiré and moiré singlet from laterally displaced non-periodic photonic slabs	Cheng, Xi; Mo, Huilin; Duan, Zhenjie et al.	2026		0	10.1038/s42005-026-02641-4
The Role of Fibonacci sequence in the Transport of Bose-Einstein Condensate Atoms in Optical Lattices	Eksioglu, Yasa; Akdeniz, K. Gediz	2007		0	10.1063/1.2733338
Dynamically confined superwires as stable electron guides in the presence of lattice defects	Kim, Myeongseo; Graf, Anton; Lin, Ke et al.	2024		0	(link)
Band Flips and Bound States in Leaky-Mode Resonant Photonic Lattices	Magnusson, Robert	2018		0	(link)
Infrared resonance-lattice device technology	Magnusson, Robert; Ko, Yeong H.; Lee, Kyu J. et al.	2024		0	10.48550/arXiv.2404.13167
Compact localized states and flatband generators in one dimension	Maimaiti, Wulayimu; Andreanov, Alexei; Park, Hee Chul et al.	2017		0	(link)
Localized Structures in Spatially Extended Systems: Fronts and Defects	Knobloch, Edgar	2019		0	(link)
Tamm states and variety of their analogs in electrodynamics	Tarapov, S. I.; Belozorov, D. P.	2013		0	10.1109/MSMW.2013.6621985
Collaborative Research: Cellular Metamaterials that Localize Stress - Towards a Topological Protection against Fracture	Gonella, Stefano; Labuz, Joseph F	2020		0	(link)
Lattice multi-lobed solitons in photorefractive materials	Salgueiro, José R.; Michinel, Humberto; Rodas-Verde, María I. et al.	2008		0	10.1088/1464-4258/10/9/095103
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From mathematical order to functional materials: new developments in thermal transport engineering	Nomura, Masahiro; Wu, Xin	2026		0	10.35848/1882-0786/ae45ec
Light-induced forces on small particles	Ng, Tsz Fai Jack	2005		0	(link)
Real-space and Real-time Study of Two-dimensional Colloidal Quasicrystals	Wu, Ning; Wu, David T	2020		0	(link)
5f Band Magnetism in Uranium Compounds	Arko, A. J.; Joyce, J. J.; Moore, D. P. et al.	2000		0	(link)
Disordered topological graphs enhancing nonlinear phenomena	Jia, Zhetao; Seclì, Matteo; Avdoshkin, Alexander et al.	2023		0	(link)
Topological Amorphous Metasurfaces Based on Electromagnetic Duality	Bisharat, D. J.; Sievenpiper, D. F.	2020		0	10.1109/Metamaterials49557.2020.92849
RUI: Disorder in Strongly-Correlated Electrons on a Lattice	Khatami, Ehsan	2016		0	(link)
Multi-Worm Algorithm for Path Integral Quantum Monte Carlo in Ultracold Dipolar Gases	Capogrosso-Sansone, Barbara	2015		0	(link)

Title	Authors	Year	Venue	Cited	DOI
Single-layer resonant metasurfaces as versatile IR sensors, polarizers, filters, and reflectors	Magnusson, Robert; Ko, Yeong H.; Lee, Kyu J. et al.	2024		0	10.1117/12.3021205
Finite quantal systems – from semiconductor quantum dots to cold atoms in traps	Reimann, Stephanie M.	2007		0	(link)
Dynamics of Time-Varying and Nonlinear Phononic Lattices	Kim, Brian Lee Kiwon	2023		0	(link)
RUI: Quantum Transport Dynamics with Ultracold Atoms: Localized versus Extended States	Das, Kunal	2010		0	(link)
COLLABORATIVE RESEARCH: DMREF: Designing Plasmonic Nanoparticle Assemblies For Active Nanoscale Temperature Control By Exploiting Near- And Far-Field Coupling	Willets, Katherine	2021		0	(link)
Localized coherent structures in the three-dimensional high-order nonlinear Schrödinger equation with quasi-periodic potentials	Wang Y.	2026	Wuli Xuebao Acta Physica Sinica	0	10.7498/aps.75.20260181
Broadband High Absorption Based on Dirac Semimetal Quasi-Periodic Structure	Zeng X.	2026	Laser and Optoelectronics Progress	0	10.3788/LOP251698
Progress in Moiré Photonics of Different Dimensions (Invited)	Huan S.	2026	Guangxue Xuebao Acta Optica Sinica	0	10.3788/AOS260904
Exploring aperiodic order in photonic time crystals	Coppolaro M.	2025	Physical Review Applied	0	10.1103/vw6g-kq3z
Non-Hermitian quasicrystal lattices	Ghatak A.	2024	CLEO Fundamental Science CLEO Fs 2024 in Proceedings CLEO 2024 Part of Conference on Lasers and Electro Optics	0	10.1364/cleo_fs.2024.ftu3g.7
Localization and delocalization in kicked quantum matter	Shimasaki T.	2022	Proceedings of SPIE the International Society for Optical Engineering	0	10.1117/12.2634143
Research Progress on Photonic Quasicrystals	Che Z.	2021	Rengong Jingti Xuebao Journal of Synthetic Crystals	0	(link)
Disorder-enhanced transport in photonic quasi-crystals: Anderson Localization and delocalization	Levi L.	2010	Optics Infobase Conference Papers	0	(link)

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openalex	656	319	37
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inspire	17	9	0
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ads	600	352	0
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2010	20
2011	11
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2014	28
2015	38
2016	78
2017	86
2018	127
2019	143
2020	142
2021	143
2022	152
2023	155
2024	236
2025	240
2026	483

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Venue	Records
arXiv preprint	345
Advanced Functional Materials	58
Advanced Energy Materials	51
arXiv (Cornell University)	50
Advanced Materials	35
ACS Applied Materials and Interfaces	35
Small	24
Chemical Engineering Journal	22
ACS Energy Letters	20
Solar RRL	19
Nature Communications	18
ACS Applied Materials & Interfaces	18
Journal of Materials Chemistry A	17
Solar Energy	17
Nano Energy	17

Most frequent authors

Author	Records
Michaël Grätzel	21
Marco Bernasconi	19
Shaik M. Zakeeruddin	15
Nam-Gyu Park	15
Omar Abou El Kheir	13
Jiangzhao Chen	11
Shu Chen	11
Wei Huang	10
Un-Gi Jong	10
Zheyong Fan	10
Chol-Jun Yu	10
Longwen Zhou	10
Shimasaki, Toshihiko	10
Dotti, Peter	10
Anders Hagfeldt	9

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Venue	Records removed
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Zenodo	11
SSRN Electronic Journal	9
Open MIND	1
Preprints.org	1

Count history

Per-block totals across archived runs (counts_history.csv); drift shows how the indexes – or your queries – changed.

Block	20260815T095908	20260828T091142	20260828T100120
PSC	-	-	14,620
MLP	-	239	838
NOV	-	0	0
QC	-	-	978

Suggestions

- Block PSC: openalex 4,567, semanticscholar 2,968, ads 2,179, scopus 4,731 hits – a generic term is probably driving this; tighten a group or add a more specific one before reading.
- Block PSC: counts differ >20x across backends (175 to 4,731); grammar and coverage diverge, so do not compare these numbers – discover here, quote one source.
- Block NOV: zero hits on every backend. Either the intersection is genuinely empty (a finding – check the synonyms first) or one group is too narrow; try dropping one group and rerunning.
- Block QC: counts differ >20x across backends (17 to 414); grammar and coverage diverge, so do not compare these numbers – discover here, quote one source.
- 4 block/backend pair(s) hit the `--limit` cap (300); raise it (largest total 4,731) if you need the complete record set rather than the most-cited slice.
- Open-access status was not looked up; rerun with `--pdfs` (optionally `--pdf-blocks`) to collect legal OA PDF links via Unpaywall.
- The PRISMA flow's manual stages are empty: fill in prisma.json (screened / excluded / assessed / included) and rerun report.py to complete the diagram.
- No journal metrics on file; run `python journals.py fetch` (OpenAlex, no key) to add impact-factor-like figures and enable `--min-metric`.
- Block MLP: total hits changed from 239 (run 20260828T091142) to 838; count drift is expected as indexes grow, but a large jump usually means the query changed – diff queries.json between the runs.